Black Gold: The Effect of Wealth on Descendants of the Enslaved*

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Abstract

This paper examines how positive wealth shocks affected the economic trajectories of descendants of the enslaved in the early twentieth-century United States. I exploit a natural experiment in which Creek Freedmen allottees—Black landholders in Oklahoma—received quasi-random windfalls when producing oil wells were drilled on their land allotments. By constructing and linking censuses and maps from the Bureau of Indian Affairs, oil drilling records, and U.S. censuses from 1910 to 1940, I show that oil discoveries were as-good-as-random with respect to core pre-windfall individual characteristics and pre-treatment outcomes. The wealth shocks had modest direct effects on asset accumulation, but large and persistent impacts on human capital. Treated youths remained in school longer—treated women were 36.4 percentage points more likely to still be enrolled at age 18—and, in adulthood, were 11.3 percentage points more likely to hold white-collar occupations, nearly triple the rate among their untreated peers. These gains extended to the next generation: children of treated mothers were 6.3 percentage points more likely to attend school. These findings provide the first causal evidence on the long-run effects of wealth shocks for descendants of the enslaved. They suggest that wealth enabled productive investments in education and mobility, generating lasting socioeconomic gains despite ongoing racial barriers. *JEL* Codes: E21, N32, D31, I38, J15.

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1 Introduction

Today, stark disparities exist between Black and white Americans on a number of dimensions, including homeownership rates, education levels, income, and wealth. One potential explanation for these disparities is that, upon emancipation, formerly enslaved Black Americans had neither wealth nor access to credit due to legal and social barriers, preventing them from forgoing short-term needs in order to make long-term investments, such as those in human capital or real estate. While a half-century of scholarship has cast doubt on the possibility that greater resources post-slavery could have made a meaningful difference to the economic positions of descendants of the enslaved, that conclusion has rested on remarkably little empirical evidence. The historiography underlying that conclusion is largely qualitative (Engerman, 1982; Foner, 1981; Higgs, 2008; Ransom, 2005; Woodman, 1977, 2001), and the small quantitative literature either compares families who accumulated land or savings with those who did not (Collins et al., 2024; Miller, 2020)—comparisons in which wealth may reflect the very family characteristics that also shape descendants' outcomes—or studies the loss of wealth rather than its receipt (Hornbeck and Keniston, 2024). What has been missing is wealth assigned quasi-randomly across otherwise similar families of the formerly enslaved.

In this paper, I leverage a natural experiment in which hundreds of emancipated people, their children, and their grandchildren quasi-randomly received large windfalls following oil discovery on their families' land, while other observationally similar people did not. This unique historical event enables me to isolate the impact of greater wealth on short and long-term Black economic progress.

Thousands of formerly enslaved people and their descendants became landowners in present-day Oklahoma during the early 1900s, due to federal Native American policy. Over the following two decades, a small fraction of these landholders profited from the unexpected and unprecedented discovery of oil on their land. Three features of this empirical setting are key to my identification strategy. First, as a result of federal "land allotment" policies, each enrolled man, woman, and child received the same amount of land: 160 acres. Second, nearly all of these land allotments were assigned to families before the discovery of oil. Third, federal oversight and prevailing industry practice standardized the lease contracts that oil speculators offered to landowners, limiting the scope for bargaining between landowners and oil companies to influence the amount of oil revenue.

I find that greater wealth enabled Black households to make productive investments in human capital, generating lasting socioeconomic gains despite ongoing racial barriers. Young landholders who benefited from oil discoveries remained in school longer than their peers who did not: treated women were 36.4 percentage points more likely to still be enrolled at age 18, an age by which barely half of untreated women remained in school. Having stayed in school longer, treated men then caught up in family formation, becoming 28.3 percentage points more likely to have married by age 24. These educational investments carried into adulthood: allottees treated in youth were 11.3 percentage points more likely to hold white-collar occupations, nearly tripling the 6 percent base rate. These gains extended to the next generation: the children of treated allottees were 5.1 percentage points more likely to attend school—6.3 percentage points when the mother was treated—on a base attendance rate of 83 percent, even though treated families had no fewer children. While recipients were more likely to live in urban areas, they were no more likely to own their homes—a pattern consistent with the fact that urban residence went hand in hand with renting. Treated families nonetheless held onto the homes they had: by 1930, they were substantially less likely than untreated allottees to have exited homeownership. Thus, the evidence suggests that treated families made deliberate choices to rent in urban areas, which provided better access to labor market and educational opportunities, without eroding the housing wealth they already held. These moves, moreover, were local ones: treated allottees were, if anything, more likely to remain in the county of their allotment and within Creek Nation territory, relocating to nearby cities such as Tulsa and Muskogee rather than leaving the region.

Taking advantage of this clean empirical setting required substantial novel data digitization and processing. I construct a rich, intergenerational database of a subset of these landholders, Creek Freedmen, along with the exact locations of the more than one million acres of land they owned.[1] I combine this dataset with public records of the dates and exact locations of oil-producing activity in Oklahoma to identify which Creek Freedmen received a wealth shock and which did not. At rates exceeding typical ones in the literature among Black Americans, I link this sample to various historical datasets, in particular the decennial censuses, in order to establish pre-treatment balance over observable characteristics and to measure causal impacts of the windfall on socioeconomic variables. Freedmen who benefited from oil discoveries are statistically indistinguishable from those who did not on core time-invariant individual characteristics and the main pre-treatment outcomes, supporting the validity of the identification strategy.

Ex-ante, it is unclear whether the long-term impacts of a large positive wealth shock to descendants of the enslaved would be positive, neutral, or even negative. Positive wealth shocks could lead to lasting improvements in socioeconomic status if the money were invested in productive ends, such as educating the direct recipients or their children, establishing a business, or funding migration to a location with stronger employment opportunities or civil rights protections– mechanisms consistent with escaping a "poverty trap," in which initial wealth and the size of the shock jointly determine whether households begin a sustained path of asset accumulation (Balboni et al., 2022). However, much scholarly work is skeptical that transfers of land or wealth to emancipated people could have had any equalizing impact, arguing that severe legal and social constraints limited the ability to transform resources into long-run gains (Engerman, 1982; Foner,

[1] In this paper, when capitalized, "Freedmen" refers to individuals once enslaved by a member of a specific group of Indigenous Nations, the "Five Civilized Tribes," and their descendants. In contrast, when not capitalized, "freedmen" is an adjective that means formerly enslaved but not necessarily individuals connected to the Five Tribes (Oklahoma Historical Society, 2007). Furthermore, I use "Creek Freedmen" as an adjective to identify those included on the Dawes Creek Freedmen Rolls or as a plural noun for the same, and "*a* Freedman" as a noun to refer to an individual. Similarly, "Creek Indian(s)" references a person's Dawes enrollment category, while "Muscogee Native" refers to Native citizens of the Muscogee (Creek) Nation, regardless of whether they were enrolled on the Dawes Rolls.

1981; Higgs, 2008; Ransom, 2005; Woodman, 1977, 2001). Moreover, positive wealth shocks may have provoked anti-Black violence and political backlash that destroyed physical capital, restricted civil rights, or threatened the safety of the beneficiary communities, potentially leaving beneficiaries worse off than if they had received nothing.[2]

This paper is the first to measure the long-term causal impact of wealth on descendants of the enslaved.[3] A large number of studies have examined the short-term impacts of wealth shocks, whether from lotteries (Cesarini et al., 2017, 2016; Golosov et al., 2021) or from randomized and quasi-experimental transfer programs (Akee et al., 2010; Banerjee et al., 2017). The subjects of these studies are typically either middle-class individuals in high-income, developed nations or poor individuals in developing countries. Long-term or intergenerational studies are less common, but examples include Aizer et al. (2016), Bleakley and Ferrie (2016)—a land lottery—and Haws et al. (2025). Relative to the populations studied in these papers, the formerly enslaved population that I study differs both in terms of initial wealth or income levels and in the extent of discrimination that they faced.

The closest paper to mine is Miller (2020), which finds that 35 years after emancipation, descendants of people who had been enslaved by Cherokee Natives, and thus received land following emancipation, had higher literacy rates and owned more capital than the descendants of similar people who had been enslaved by white Americans and who never received the infamously promised "forty acres and a mule." While similar in spirit to Miller (2020), my paper does not compare landholders to non-landholders. The individuals that I compare *all* own land, and in the aggregate, differ *only* on whether they received a positive wealth shock or not.

The early Oklahoma oil boom was characterized by many of the advantages shared by the lottery and randomized control trial (RCT) studies mentioned above: The discovery of oil generated unexpected positive shocks to allottee wealth that were orthogonal to the recipients' underlying characteristics. Windfalls could

[2] Chyn et al. (2024) find a geographic relationship between Freedmen's Bureau locations (and their resulting positive impact on Black economic status) and racial violence perpetrated by white Southerners. Derenoncourt (2022) finds that negative reactions by northern communities reduced social and economic gains of Black migrants from the South during the Great Migration (1940-1970). Logan (2023) finds a relationship between tax revenue and violence against Black politicians at the county level during Reconstruction. Lastly, Althoff and Reichardt (2024) argue that persistence of racial disparities in education, income, and wealth stem largely from differential exposure to Jim Crow laws: descendants of the enslaved were more likely to remain in the South, where these policies were most aggressively enforced. Segregated schooling—mandated by laws that created separate and unequal educational systems for Black children—played a major role in reinforcing these long-term inequalities.

[3] A small number of papers have examined this question descriptively. Collins et al. (2024) demonstrate that the adult sons of emancipated fathers who had owned land in 1880 had higher rates of literacy and homeownership in 1900, with more land being associated with more literacy/homeownership. In contrast, Hornbeck and Keniston (2024) find that descendants of emancipated people who were depositors in the Freedmen's Bank and who suffered a large *negative* loss to their wealth upon the bank's closure in 1874 are statistically indistinguishable on a number of outcomes from descendants of similar emancipated people who did not suffer such a loss.

be large, and in regards to the Creek Freedmen population, their usage was largely unrestricted. However, this context offers several advantages over the lottery and RCT studies. First, when most Freedmen applied for an allotment and selected the land, they were unaware that their allotment might be associated with a bonus financial windfall. Unlike studies based on lottery participants, which must address concerns about external validity to non-lottery populations, this context offers an advantage: allottees did not anticipate the potential rewards associated with oil discovery. Second, I can follow the family of the treated allottees for at least 50 years after the original positive shock was received, whereas contemporary studies focus only on short-term impact. However, the biggest distinction between this context and most RCTs and lottery studies is the population examined: American descendants of the enslaved. Results from this population are important to our understanding of continued racial disparities in the US today.

My findings underscore that the effects of wealth take time—often decades—to fully materialize, reinforcing a growing body of research that emphasizes the importance of historical evidence in understanding the full impacts of welfare policies (Aizer et al., 2024, 2016). They also caution against generalizing results from existing studies of wealth shocks, which largely draw on samples of white households in developed nations or poor households in developing countries. The starting conditions for descendants of the enslaved in the United States were fundamentally different: They faced political disenfranchisement, legal discrimination, and financial exclusion under Jim Crow. Moreover, the magnitude of oil wealth shocks relative to recipient household income–63% of annual household income at the median–is substantially larger than the magnitude of many of those studied, lending credence to the efficacy of "big push" type policies studied in developing countries, in contrast to the examples of smaller income transfers typically found in the United States.

The remainder of the paper is organized as follows. Section 2 describes the historical events leading up to the natural experiment in this paper. Section 3 describes the construction of the novel datasets used in this study, provides summary statistics, and establishes balance over many pertinent characteristics. Section 4 lays out my estimation strategy and introduces the three estimation samples—Older Allottees, Younger Allottees, and Children of Allottees—used throughout the paper. Section 5 presents my results for Younger Allottees, focusing on school enrollment and later occupational upgrading. Section 6 examines effects on the next generation, showing that oil wealth raised children's school attendance without reducing family size. Section 7 reports effects on occupational status, urbanization, and homeownership. Section 8 shows that these gains were achieved in place: treated allottees remained near their allotments and did not sort into observably different neighborhoods. Section 9 concludes.

2 A Brief History of the Muscogee (Creek) Nation

2.1 The Muscogee (Creek) Nation Prior to Allotment

During the 17th and 18th centuries, several Indigenous villages in present-day Georgia and Alabama coalesced into the "Muscogee Confederacy" or the "Creeks." While they mostly practiced communal agriculture at first, by the second half of the eighteenth century, some Muscogee farmers adopted slavery and engaged in plantation agriculture, as practiced by their white neighbors. Because of this move toward assimilation with Anglo-American norms, as well as other practices, the Muscogee and four other Indigenous nations earned the dubious distinction as the "Five Civilized Tribes," in contrast to the remainder of "...other so-called 'wild' Indians who continued to rely on hunting for survival" (Frank, 2007).

The Muscogee's efforts toward assimilation did not shield them from the Removal Act of 1830, federal law that forced all Native Americans living east of the Mississippi River to move to an "Indian Territory," west of the Mississippi. The Muscogee people were forcibly moved west over the ensuing decade, bringing their enslaved people with them. In Indian Territory, in the decades leading up to the Civil War, usage of private land and the practice of plantation slavery intensified among the Muscogee. Following emancipation in 1865, the Muscogee (Creek) Nation recognized the freedom and citizenship of the formerly enslaved, who comprised 13% of the population at that time, and granted them land-use rights commensurate with those held by Muscogee Natives (Chang, 2010, p.33).

In an effort to weaken the political power of Native nations and open Native reservation land to white settlement, Congress passed the General Allotment Act of 1887, also known as the Dawes Act, which broke up communal Native land into privately owned parcels known as allotments. The Five Civilized Tribes, including the Muscogee (Creek) Nation, successfully resisted inclusion in the Dawes Act at that time, but eventually ceded under federal pressure, agreeing to undergo allotment under the Curtis Act of 1898 (Kidwell, 2007). The agreement between the United States government and the Muscogee stipulated that every Muscogee citizen was entitled to 160 acres of land, regardless of age, gender, or former enslavement status.

2.2 The Muscogee (Creek) Nation under Allotment

In April 1899, the Dawes Commission began to enroll applicants onto the "Dawes Rolls," formally designating them as members of the Muscogee (Creek) Nation.[4] Successful applicants had to demonstrate either Creek lineage to be enrolled as "Creek Indians," or that their ancestors were enslaved within the Muscogee (Creek) Nation prior to the Civil War to be enrolled as "Creek Freedmen." The Dawes Rolls closed

[4] In fact, having ancestors listed on the Dawes Rolls remains a requirement for citizenship in the Muscogee (Creek) Nation today (The Muscogee Nation, 2023). A recent court ruling affirmed that descendants of Creek Freedmen are entitled to citizenship (Press, 2023).

for the first time in September 1904 but reopened twice to enroll infants born after the original deadline, finally closing for good in March 1907. For more details and excerpts of text from the agreements, see Appendix Subsection E.2. Inclusion on the Dawes Rolls entitled individuals–or their parents or guardians–to choose an allotment of 160 acres of land from the Muscogee (Creek) Nation.[5] Allottees typically selected land on which they and their family already lived or farmed (Chang, 2010, p. 76).[6] The vast majority of eventual allottees had selected their allotments by September 1901, and deeds began to be issued in December 1901 (Chang, 2010, p. 62).

At the turn of the twentieth century, the land near the northern boundary of the Muscogee (Creek) Nation– later the site of Tulsa, Oklahoma–lay within a much larger expanse of Muscogee (Creek) Nation (see Appendix **Figure I.6**). Though perhaps intended to transform the Muscogee into yeoman farmers, the federal government's allotment policy quickly produced a landscape characterized by landlords and tenants (Chang, 2010, p. 72). I describe the primary reasons for this change in the next subsection.

2.3 Oil Discovery in the Muscogee (Creek) Nation

In late summer 1913, 11-year-old Sarah Rector was on track to become "the Richest Black Girl in America" (Bolden, 2014). How the middle daughter of an Oklahoma farming family, descended from enslaved people, was reportedly bringing in $300 a day– equivalent to nearly $10,000 in 2025– was a matter of chance. As her grandparents were enslaved by Muscogee Natives, her parents Joe and Rose enrolled the family with the Dawes Commission, first themselves in 1899, then her older sister Rebecca in 1902, and lastly Sarah and her little brother Joe Jr. in 1905 as "Newborn Freedmen." The enrollment entitled each family member to select an allotment of 160 acres of land in the Muscogee (Creek) Nation, and when Oklahoma became a state in 1907, the family was living on and farming Rose's allotment in Muskogee County. However, it was Sarah's allotment, over in Creek County, that would later change the direction of the family's fortunes.

The first discovery of oil in the Tulsa area occurred in 1901 on an allotment belonging to Creek Indian Sue A. Bland. While the well itself produced minimal oil, this discovery is considered to have incited the Tulsa oil boom (Weaver, 2007). The first "gusher" oil well, named for its geyser-like appearance, was drilled in 1905 on the allotment of another Creek Indian, Ida Glenn, after which the lucrative "Glenn Pool" was named. Appendix **Figure I.5** presents a 1912 map of Glenn Pool. Soon after, investors and speculators

[5] Of the 160 acre allotment, 40 acres would be designated as their "homestead" and would be "nontaxable and inalienable and free from any encumbrance whatever for twenty-one years." The remaining 120 acres, known as the "surplus," were only covered by these restrictions for the first five years. Federal treatment toward Creek Freedmen soon began to diverge from treatment toward Creek Natives and, in April 1904, Congress removed restrictions on surpluses held by adult Freedmen. In 1908, it removed restrictions on their homesteads too, as well as all restrictions on homesteads and surpluses of Freedmen minors. See Appendix Subsection E.1.

[6] Qualitatively, there is little evidence of land selection with respect to future oil earnings. I address this concern quantitatively in Section 3.2.

flooded the area and made exploratory contracts with many landholders for the right to drill for oil on their land.[7]

Two of those exploratory contracts were made by "landman" Frank Barnes with Sarah Rector's father in March 1912 for the exclusive right to drill on Sarah's and Joe Jr.'s land. Just for signing each exploratory contract, the Rectors received $80, or fifty cents an acre. Barnes soon sold the leases to a driller named B.B. Jones, who in August 1913 drilled a well on Sarah's land and discovered a gusher that output 2,500 barrels of oil a day.

Over the next three decades, drillers found oil under many allotments owned by Muscogee Natives and Freedmen (see Appendix **Figure I.7**).[8] **Figure 3** shows a histogram of the distribution of the date that drilling was reported as complete on each unique well located in the Muscogee (Creek) Nation (oil-producing, dry, gas, or otherwise). Drilling ramped up precipitously between 1910 and 1920, peaking in this area in the mid 1920s.

The owners of allotments with producing oil wells received payments, sometimes substantial, via royalties from the oil company that drilled.[9] I estimate that the average allottee who held a lease with an oil company by 1909 earned $257 that year, a sizable windfall, considering that the typical American farm family earned only about $402 that year (Goldenweiser, 1916). The distribution of earnings from royalty accounts over the years 1908-1918 can be found at Appendix **Table I.10**.

Not much more is known about the later life of the most famous, but exceedingly private, Creek Freedmen allottee, Sarah Rector, and her family following the oil strike on Sarah's land. The daughters spent time at the Tuskegee Institute in Alabama during their childhood, and eventually, the family moved to Kansas City, Missouri (Sar, 1914, 1920).[10]

[7]For more details on oil drilling contracts, including a sample notice of regulation changes, see Appendix Section F.

[8]One might wonder whether Freedmen differ from Muscogee Natives in likelihood of oil discovery. Such a disparity could occur in multiple plausible ways: if Natives and Freedmen faced different institutions, had different ownership rights and restrictions, or simply selected land of different quality due to Muscogee Nation or federal government discrimination. I show in **Table I.18** that I do not observe differential rates of oil discovery over Dawes enrollment category (Creek Freedmen versus Creek Indians).

[9]One might wonder whether the royalty, or "mineral rights owner's share," that oil companies paid Freedmen allottees, typically 1/8th (12.5%), is a lower rate than their white counterparts received at the time. However, this royalty was standard from 1900 through the 1980s (Oklahoma Mineral Rights and Royalty, nd). Given the degree of oversight that the Bureau of Indian Affairs had over the business affairs of allottees at that time, it is not surprising that Indian and Freedmen received comparable or *better* agreements than their white counterparts.

[10]Newspaper accounts of the pursuits of select other individuals who grew up in households treated with oil can be found at Appendix Section C.

3 Data

Because I am interested in comparing the outcomes of descendants of the enslaved who received a wealth shock from oil discovery (my treatment group) and those who did not (the untreated group), I compile a dataset of individuals who were allotted land in the Muscogee (Creek) Nation and identify which allotments yielded oil. Tracking these individuals, along with their immediate family members and grandchildren, across pre-shock and post-shock data sources allows me to verify that treated individuals were comparable to untreated individuals and to analyze their later outcomes.

3.1 Data Sources and Construction

I start by building a database of allottees for the entire Muscogee (Creek) Nation that contains limited, time-invariant microdata on all allottees, such as age reported at date of enrollment, gender, and the names of their mother and father and which Indigenous nations they were enrolled in, if any. I also observe a number of reported dates of death. The dataset contains 18,850 allottees in total, but I am only interested in Freedmen allottees, or 6,839 individuals. More information on the Creek allottee dataset and a full list of the variables can be found in Appendix Subsection H.1.

Half of the Freedmen allottees were female (**Table I.17**). In 1910, Freedmen were 25.3 years old on average and half were aged 22 or younger. However, the population spanned ages four to ninety, as shown in **Figure 4**. The maximum possible number of allottees living at any point in time between 1900 and 1915 is represented by the blue area in **Figure 5**. While the full Creek Freedmen population may have been 6,839, a smaller number was alive at any point in time, as, for example, 1,583 people (23%) were born between 1900 and 1907, and 855 (12.5%) are indicated to have died by 1910.

I next determined the exact location of every allotment in the Muscogee (Creek) Nation.[11] I describe this process in Appendix Subsection H.4.

Lastly, I obtain a dataset of every "drilling action" taken in Oklahoma from the Oklahoma Corporation Commission (OCC). I focus on oil-producing wells that were drilled within the boundaries of the Muscogee (Creek) Nation by the end of 1918. See Appendix Subsection H.5 for more details on this dataset.

Treatment Assignment

An allotment is considered to have been treated if any producing oil wells were built on the land by the end of 1918. As shown in **Figure I.12**, I assign treatment status by merging the three datasets described above: I first link the allotment land with the oil well data by simply merging the PLSS codes of the well

[11] I use the word "allotment" to refer to the entire 160-acre entitlement of one individual. In the case where an individual's allotment is divided into multiple non-contiguous segments, I still refer to the sum of the land owned by that particular allottee as "allotment," singular.

actions "many-to-one" to the PLSS codes of land allotments.[12] I then link each treated allotment with its allottee owner via the allottee's unique roll number.

Then, for each Freedmen allottee, I can observe whether their allotment produced oil by the end of 1918 (treatment) or did not (control).[13] For Freedmen defined as treated under this definition, the median first year of oil production is 1916. I plot allotments by the first year that oil was produced on them in **Figure 6**. The progress of the treated and untreated sample sizes over five-year intervals can be seen in **Table 1**, which shows that the arrival of new oil-producing allotments peaked around 1920 and fell steadily afterward, with oil having been discovered under the allotments of 446 different people by 1920. Oil was discovered under the allotments of 293 different people by the end of 1918, which I currently use as my conservative cutoff of treatment receipt. By this definition, the remaining 6,546 allottees make up the control group.

Record Linking

While, as described above, the Creek Freedmen database includes a small number of variables for all allottees, in order to speak to any other pre-treatment observables, it is necessary to link individuals to external records, in particular to the 1910 census.[14] Then, in order to study outcomes, I need to link my sample to records collected in the post-treatment years. This paper presents estimates using information from the 1920, 1930, and 1940 censuses. Future work will incorporate the 1950 census.

The empirical strategy that I describe in Section 4 does not strictly require a panel in order to draw causal inference. Given the unique characteristics of my data, I design my own linking methodology, and first conduct each record linkage *separately* from every other record linkage. I supplement the sample that I personally linked with pre-made links, made by various census-linking projects, keeping their various strengths and weaknesses in mind and choosing the conservative options when appropriate (Abramitzky et al., 2021; Bailey et al., 2020; Buckles et al., 2023). This process is depicted in **Figure I.3**. To be clear, these projects link the decennial censuses *to each other* and my biggest challenge is linking an *external data source*–the Dawes Rolls–to any decennial census. See Appendix Section D for more details.

Table I.15 shows that there are differential linkage rates by observable characteristics, but they occur

[12] By taking advantage of the PLSS, I avoid needing a geographic information system (GIS) software.

[13] I define ever/never treated as "oil by the end of 1918" or "oil after 1918 or never at all." Creek Freedmen originally were prohibited from selling the entirety of their allotment for twenty-one years after 1901, just the same as Creek Indians. However, those restrictions were lifted incrementally over the following three decades, with difficulty of land sale differing with respect to race, age and whether the land was designated as "homestead" or "surplus." See Appendix Section E.1. Contemporary accounts suggest that many Creek Freedmen sold or otherwise lost their land relatively quickly. The royalty-account records summarized in Appendix **Table I.10** nonetheless document allottees receiving oil income through 1918.

[14] While it would be ideal to observe this population in 1900, prior to the arrival of *any treatment whatsoever*, and though the Indian Territory was enumerated during the 1900 census, unfortunately the records have not been made available as computer-readable data.

in predictable ways that we would not expect to be related to oil discovery.[15] The matched sample is younger and more male.

3.2 Summary Statistics and Balancing Tests

To confidently analyze later-life outcomes of oil discoverers as resulting from a natural experiment, I first establish that "treatment assignment" can be treated as random. That is, receiving a wealth shock was uncorrelated with pre-treatment observable characteristics.

As I describe in Subsection 3.1, I observe the age and gender of all units in the full Creek Freedmen allottee sample. This microdata can be considered "ground truth." **Table H.1**, Panel A, demonstrates that treatment and control units are statistically indistinguishable on these time-invariant characteristics: likelihood of receiving treatment does not vary by gender or age.[16] The average treatment and control allottee enrolled on the Dawes Rolls in late fall, 1899. Radically different enrollment dates among treatment and control allottees, in either direction, earlier or later, would be cause for concern: it could imply asymmetric oil distribution information between the treatment group and control group.

In addition, among those matched to the 1910 Census, treatment and control units were also similar on time-invariant census observables, with the possible exception of mother's birthplace. This can be seen in Panel B.[17] A joint F-test of the predetermined characteristics in Panel B yields $F(6, 917) = 1.99$ with a p-value of 0.065, so I cannot reject the null hypothesis that these covariates are jointly unrelated to oil discovery at the five-percent level. It is unsurprising that matched individuals are not totally representative of the full population, whose true ages and genders we know from the Dawes Rolls. The people matched to the census may be slightly younger.

Lastly, **Table H.1** demonstrates that in 1910, treatment and control units are also statistically indistinguishable over the variables that I will later analyze as outcomes during the post-treatment decades.

4 Estimation Strategy

The central challenge in estimating the effect of wealth on descendants of the enslaved is the endogeneity of wealth. Freedmen and their children who have greater wealth than others likely differ in other ways that make it difficult to interpret subsequent differences in other characteristics, such as educational attainment, as resulting directly from the wealth itself. In this paper, I overcome that challenge by demonstrating that the receipt of wealth shocks by some people but not others is exogenous to other characteristics.

[15] See differential link rates by oil discovery at **Table I.14**.

[16] Likelihood of treatment also does not vary between Creek Indians and Creek Freedmen, as shown in Appendix **Table I.18**.

[17] The treatment and control units that make it into the three post-treatment samples–those matched to the 1920, 1930, and/or 1940 censuses respectively–are also balanced on time-invariant characteristics, as demonstrated in **Table I.16**. The one exception is gender. Many more women in the treatment group were matched to the 1930 and 1940 censuses than women in the control group.

The treatment effect of finding oil can be assessed directly by comparing mean outcomes for finders and non-finders, or by estimating a simple bivariate regression with an outcome on the left-hand side and a dummy variable for finding oil on the right-hand side. Because oil discovery is as-good-as-random with respect to allottee characteristics (Section 3.2), omitted-variable bias is not a first-order concern and controls are not strictly necessary to recover the effect of treatment. I nonetheless adopt the regression-based approach, which permits the inclusion of a small set of predetermined controls that improve the precision of the estimated treatment effect by reducing residual variation. I estimate OLS regressions of the following form:

$$Y_{ijt} = \gamma T_j + \boldsymbol{\beta}\mathbf{X}_i + \lambda_t + \epsilon_{ijt}, \tag{1}$$

where i is the individual, j indexes the allotment, and t indexes the census decade (1920, 1930, or 1940) in which the outcome is observed. T_j denotes treatment—a producing oil well on allotment j by the end of 1918—which is a binary variable. The vector of predetermined controls $\mathbf{X}_i$ contains an indicator for gender and a quadratic in age. Specifications that pool outcomes across census decades include census-year fixed effects, λ_t; single-decade cross sections omit them. When the outcome belongs to the allottee herself, $i = j$. When I examine the outcomes of allottees' children, a single allotment j can generate multiple observations i.

I estimate each specification separately for three samples of Creek Freedmen, each defined by cohort and relationship to the allotment rather than by treatment status; every sample contains both treated and untreated individuals. *Older Allottees* were born in 1892 or earlier and were adults by the time oil arrived; *Younger Allottees* were born after 1892 and were minors at the time. *Children of Allottees* are unenrolled sons and daughters who hold no allotment of their own and are treated, if at all, through a parent's producing well. Each sample therefore carries its own control group: untreated Older Allottees, untreated Younger Allottees, and the children of untreated allottees, respectively. The two allottee cohorts are described further in Data Appendix H.1, and the Children of Allottees sample in Data Appendix H.2. Results are reported separately for Older Allottees and Younger Allottees, and for the pooled allottee sample; results for Children of Allottees appear in Section 6.

The censuses do not record wealth directly for this period, and only record income in 1940. Throughout the paper, I therefore use occupational status as a proxy for income and homeownership as a proxy for wealth; educational attainment, measured while treated allottees were of school age, is the third headline outcome. Urban residence enters the analysis not as an outcome in its own right but as a channel: Section 7 tests formally whether the move to cities mediates the relationship between oil wealth and homeownership.

Standard errors are clustered at the family level. The coefficient of interest, γ, measures the causal effect of oil discovery on the outcome.

When the outcome is binary, equation (1) is a linear probability model, and γ has the convenient interpretation of a percentage-point change in the probability of the outcome. One family of outcomes, however, is polytomous rather than binary: a worker's occupational class is one of three mutually exclusive and exhaustive categories—white-collar, blue-collar, or farming. Estimating equation (1) separately for each category preserves the percentage-point interpretation but ignores the constraint that the three probabilities sum to one. I therefore complement the linear probability estimates with a multinomial logit that models the three occupational choices jointly. Taking blue-collar employment as the base category, I estimate the following pair of equations:

$$\ln\left[\frac{\Pr(Y_{ijt}=m)}{\Pr(Y_{ijt}=\text{blue-collar})}\right]=\alpha_m+\gamma_m T_j+\boldsymbol{\beta}_m\mathbf{X}_i+\lambda_{mt}, \tag{2}$$

for $m \in \{\text{white-collar}, \text{farming}\}$, with the same treatment variable, controls, and census-year fixed effects as in equation (1). In this framework, γ_m is interpreted as the change in the log odds of being employed in occupational class m, relative to blue-collar employment, associated with treatment. I report cluster-robust standard errors, clustered at the family level. The multinomial logit results appear in **Table 5**; Section 7 compares them to the linear probability estimates.

Finally, one outcome—the distance of an allottee's census-decade residence from the county of their allotment—is an ordered categorical variable. For that outcome I supplement the linear probability estimates with an ordered logit, presented in Appendix **Table I.3** and discussed in Section 8.

5 Staying in School: The Windfall and the Transition to Adulthood

If the path to greater Black wealth long term is through higher earnings, which Derenoncourt et al. (2023) would frame as a "flow value," then investment into education would be prudent if it opens the door to professional, typically higher paying, jobs. This section covers Younger Allottees—Creek Freedmen born after 1892, minors when oil arrived—using own-allotment oil treatment. I will consider the possibility that oil money allowed families to invest more in human capital–whether that investment was in the form of school fees and supplies or keeping teenagers in school rather than employing them on the family farm–and that human capital investment allowed allottees to enter higher-class, higher-paying occupations as adults.

Table 2 traces school enrollment across the ages at which school-leaving decisions are made, using enrollment indicators constructed by monotone inference from census records (see the table note). The pattern is exactly what a relaxed resource constraint would predict: effects are negligible where enrollment is near-universal and grow precisely where staying in school becomes a genuine economic choice. At ages 12 and 14, when essentially every allottee is enrolled, there is no room for an effect: every treated allottee observed at those ages is in school, as are 98 to 100 percent of the untreated. The effects then emerge at the school-leaving margin: at age 16, 35 of the 36 treated allottees remained enrolled, against 88 percent of the untreated, and treated allottees are 23.6 percentage points more likely to be enrolled at age 18,[18] an age by

[18] At ages where enrollment among treated allottees is at or near 100 percent, the outcome has little or no variation in the treated group and cluster-robust standard errors are unreliable at this boundary. Coefficients for cells with no treated variation at all are

which only 57 percent of allottees remained in school. Beyond the school-leaving ages, where enrollment among Creek Freedmen falls to 18 percent at age 20 and 8 percent at age 22, the estimates lose precision: the age-20 point estimate remains large at 25.9 percentage points but is significant only at the ten-percent level, and the age-22 estimate (14.4 percentage points) cannot be distinguished from zero. These patterns are not an artifact of any single source: Appendix **Table I.4** decomposes the estimates by data source, and census enrollment records alone (24.7 percentage points at age 18) and educational attainment reported in the 1940 census alone (22.5 percentage points at ten years of schooling completed, 25.8 at fourteen) independently confirm that treated youths stayed in school longer. Oil money kept teenagers in school through exactly the years when their untreated peers were leaving.[19]

The right panel of **Table 2** shows that this extended schooling carried into adulthood as occupational upgrading. Among Younger Allottees observed with an occupation in the census, oil raises the probability of white-collar employment by 11.3 percentage points, nearly tripling the 6 percent base rate, and lowers blue-collar employment by 10.3 percentage points. The effect on farming is a precisely estimated zero (-1.0 percentage point on a base of 24 percent): the upgrade operated as a movement out of wage labor and into white-collar work rather than an exodus from the land. Section 7 shows that this estimate is robust to modeling the three occupational classes jointly.

6 The Next Generation: Child Investment without a Fertility Tradeoff

This section considers three samples in turn: allottees' own completed fertility (**Table 3**, Panel A, own-allotment treatment); the Children of Allottees sample—unenrolled sons and daughters treated through a parent's allotment (**Table 3**, Panel B); and allottees transitioning from school to work and household formation (**Table 4**, own-allotment treatment).

The canonical quantity-quality tradeoff theory of fertility proposes that families face a tradeoff between the number of children to have (quantity) and the investment to make in each child's upbringing (quality) (Becker and Lewis, 1973). The theory suggests that as families have more children, the resources available to devote to each child decrease, potentially leading to a lower "quality" of upbringing per child. On the other hand, if a family has fewer children, they can allocate more resources per child, providing a better "quality" of upbringing. This model can be extended to include potential income from child labor, particularly in

suppressed, shown as dashes in the tables. For near-boundary cells, Fisher's exact test on the treatment-by-enrollment table provides a conservative benchmark: it yields $p = 0.11$ at age 16, while the age-18 and age-20 estimates remain significant under exact inference, with p-values of 0.03 and 0.02.

[19] Completed secondary and higher education, collected only in the 1940 census, are not featured as outcomes in the main text because the sample is too small to detect plausible effects. The scope for such effects was narrow to begin with: nationwide in 1940, only 12.3 and 1.6 percent of Black American adults held high school and college diplomas, respectively (National Center for Education Statistics, 2017).

the context of agrarian societies, where the immediate economic benefits of an extra hand on the farm can outweigh the long-term benefits of investing heavily in each child's education and health (Barro and Becker, 1989; Boserup, 2013; Rosenzweig and Wolpin, 1980; Schultz, 1988).

If child quantity-quality tradeoff dynamics are at play in this context, we might expect that as Creek Freedmen move out of farming and into professional employment, they would bear fewer children but increase their investment in the human capital of the children that they do have. On the quantity margin, **Table 3** shows that oil discovery did not change completed fertility as measured by offspring linked in census records. Among allottees followed to age 50, the pooled effect on offspring count is 0.209 against a mean of 3.4 children, the estimate by age 40 is 0.122 and both estimates are indistinguishable from zero ; neither the women's nor the men's estimates are statistically significant at any horizon.

The adjustment instead appears on the investment margin, at precisely the ages when school-leaving decisions are made. **Table 4** follows the youngest allottees—those who were enrolled on the Dawes Rolls as infants between 1892 and 1906, and therefore treated through producing wells on their own allotments—through the transition from school to work and household formation. At age 16, every one of the 19 treated men observed at that age was still enrolled in school, against 89 percent of untreated men; with treated enrollment at the boundary, this comparison is informative about direction but cannot be precisely estimated (see footnote 18). The clearest estimate comes from women at age 18: treated women are 36.4 percentage points more likely to still be enrolled, an age by which only 55 percent of their untreated counterparts remained in school. At those same ages, no treated teenager of either sex appears in the census as a household head or spouse, against untreated rates of at most 3 percent—a byproduct of extended schooling rather than an independent finding, since a teenager who remains enrolled mechanically delays forming a household of their own. The marriage results tell the same story of delay rather than divergence: no treated man observed at age 18 had yet married, against 2 percent of untreated men, but treated men are 28.3 percentage points *more* likely to have married by age 24, the catch-up pattern one would expect if oil income allowed young people to finish school before starting families.

Read together, the two panels of **Table 3** indicate that the tradeoff operated entirely through investment rather than family size. Treated parents did not have fewer children—the offspring-count estimates in Panel A are null at every horizon—but the children they did have were 5.1 percentage points more likely to be attending school (Panel B, pooled estimate), against a baseline attendance rate of 83 percent. Unlike the allottee teenagers in **Table 4**, these are *all* children of allottees observed in the census, most born after enrollment closed in 1906 and thus holding no allotment of their own; any effect operates through parental oil wealth rather than the child's own land. Notably, the effect runs entirely through mothers: children of treated women are 6.3 percentage points more likely to be in school, while the estimate for treated fathers is negative and statistically insignificant (-0.029), suggesting that oil income accruing to women translated more directly into investment in children. Rather than trading quantity for quality, oil income appears to have relaxed the resource constraint that forces the tradeoff in the first place, allowing families to raise investment per child without reducing the number of children.

7 From Schooling to Status: Occupations, Cities, and Homes

This section reports occupational and homeownership results for the full allottee sample—and examines urbanization as the channel connecting the two—estimated separately for Older and Younger Allottees, as well as pooled, using own-allotment oil treatment throughout.

7.1 Occupations

As discussed above, I estimate the effect of oil on each occupational outcome separately using a linear probability model, which has the advantage of producing directly interpretable estimates of the change in probability associated with oil exposure. The linear probability model does, however, treat each outcome as an independent binary regression, ignoring the fact that the three occupational categories are mutually exclusive and exhaustive. The multinomial logit model imposes this restriction by construction, making it the theoretically preferred estimator for polytomous outcomes. If I employ a multinomial logit model that jointly models the probability of white-collar, blue-collar, and farming employment, using blue-collar as the base category, my main findings are preserved. These results are presented in **Table 5**. The linear probability model (**Table 2**) estimates that oil exposure raises the probability of white-collar employment by 11.3 percentage points, with no significant effect on farming. The multinomial logit's average marginal effect is nearly identical: an 11.4 percentage-point increase in the probability of white-collar employment with no effect on farming. The two estimators agree closely on both the magnitude and statistical significance of the occupational upgrading documented above. Oil had no detectable effect on the self-employment margin (Appendix **Table I.1**).

7.2 Cities and Homes

Homeownership is the main proxy for wealth available in the censuses. **Table 7** shows that treated and non-treated households have a similar propensity to own their homes, even though **Table 6** shows that treated people are much more likely to live in an urban area, where ownership rates are far lower. This urban shift operated through migration: treated allottees were 15.1 percentage points less likely to remain in their 1910 census township, and the urban effect is concentrated among those who moved (17.9 percentage points) rather than those who stayed. Treated people were also better able to hold onto their homes: the transitions analysis in Appendix **Table I.2** shows that by 1930 they were 12.7 percentage points less likely to have exited homeownership, with imprecise 1940 estimates pointing in the same direction.[20]

A robust literature in the fields of urban and development economics explores the relationship between urbanicity and poverty, with respect to segregated Black Americans and people in developing nations respectively (Kain, 1968; Lagakos et al., 2023). While Creek Freedmen may have already owned farmland and homes in rural Oklahoma, perhaps movement to a city presented employment and education opportunities that were more lucrative in the long-term, despite the large upfront migration costs of travel, rent, higher consumer prices, and loss of community. For reference, Appendix **Table I.12** presents relevant summary statistics of all of the neighborhoods included in this study, split by urban and rural status. Urban neighborhoods have twice the rate of professional occupations among nonwhite people as rural neighborhoods, in all census decades. The children in urban neighborhoods were significantly more literate than the children in rural neighborhoods, perhaps as a result of higher quality schooling opportunities and fewer household responsibilities (i.e. farming). However, it seems that the availability of higher status jobs and greater educational opportunity in cities is accompanied with lower property ownership, descriptively. Taken together, **Table 7** and Appendix **Table I.12**, suggest that in the near-term, upon receipt of oil, Freedmen chose to forego property owned in rural areas in order to make investments in relocating to towns and cities within the region—Tulsa and Muskogee, which remained inside Creek Nation territory. **Table 8** formalizes this interpretation as a mediation analysis: oil raised urban residence by 13.5 percentage points among Younger Allottees, and urban residence itself sharply depresses ownership—by 11.5 and 6.6 percentage points for the older and younger groups, respectively—so that oil's null total effect on ownership is consistent with treated families foregoing rural property to relocate, and often rent, in the cities.

8 Gains in Place: What the Windfall Did Not Change

This section examines the pooled allottee sample—Older and Younger Allottees combined—using own-allotment oil treatment, covering geographic mobility (**Table 9**) and neighborhood quality (**Table 10**).

In **Table 9**, I consider where Freedmen lived in the decades following the oil shock. Three binary outcomes capture geographic dispersion from the allotment at increasing scales: leaving the county in which one's allotment was located, leaving the boundaries of the Creek Nation, and leaving Oklahoma. The estimates pool the 1920, 1930, and 1940 censuses with year fixed effects and cluster-robust standard errors.

Column (1) presents the baseline estimates. Oil reduces the probability of being observed outside the allotment county by 7.7 percentage points, significant at the ten-percent level, relative to a sample mean of 67 percent. It similarly reduces the probability of leaving Creek Nation territory by 6.3 percentage points, also significant at the ten-percent level, against a mean of 25 percent. There is no significant effect on the

[20] House value was not collected in the 1920 census and was collected only for non-farm homes in the 1930 census. Regressions of owner-occupied house value on treatment in 1930, 1940, and pooled across years yield positive but imprecisely estimated effects.

probability of leaving Oklahoma entirely. Because these outcomes are measured relative to the location of the allotment rather than the allottee's prior residence, they capture attachment to the land rather than residential mobility per se; indeed, **Table 6** shows that treated allottees were *more* likely to change townships between 1910 and 1920. Taken together, the results indicate that the moves treated Freedmen made were local: relative to their counterparts without oil, they were more likely to remain in the county of their allotment and within Creek Nation territory.

Column (2) adds fixed effects for the county in which one's allotment was located to assess whether these patterns stem from oil wealth or simply reflect underlying migration propensities across counties. Allotment-county dummies explain only two percent of the variation in oil status, confirming that oil varied primarily at the individual level rather than the county level; nonetheless, absorbing county fixed effects controls for any persistent differences in local labor markets, mobility norms, or community ties correlated with oil incidence. Within counties, the oil coefficients attenuate and fall below conventional significance thresholds, though the negative point estimates are preserved throughout. Appendix **Table I.3** presents an ordered logit model as an alternative to the linear probability models in Table **Table 9**. The outcome is an ordered categorical variable with four levels: remaining in the allotment county, leaving the county but remaining within the Creek Nation, leaving the Creek Nation but remaining in Oklahoma, and leaving Oklahoma. Under the proportional odds assumption, the model yields a single oil coefficient summarizing the effect across all three geographic margins. Conditional on allotment-county fixed effects, the estimated coefficient is negative but not statistically significant at conventional levels, consistent with the attenuation observed in column (2) of Table **Table 9** once allotment-county fixed effects are included.

Table 10 compares the characteristics of the neighborhoods where treated and untreated Creek Freedmen allottees lived in 1940. On most dimensions, treated allottees did not live in observably different neighborhoods. Their neighbors were no more likely to have graduated college (0.037) or high school (0.028), and local schools were of similar quality as proxied by teacher salary ($62.34 on a mean of $938). Labor-market conditions were likewise indistinguishable: the estimates for the employment-to-population ratio (−0.007), the unemployment rate (0.008), and the poverty rate (−0.001) are all near zero and statistically insignificant. Treated allottees were no more likely to live in urban neighborhoods or in places with fewer single-parent families, and if anything lived among *fewer* white neighbors (−7.4 percentage points, significant at the ten percent level)—the opposite of what sorting into wealthier, whiter areas would predict. The one exception is school attendance: children in the neighborhoods where treated allottees lived attended school at a rate 1.5 percentage points higher, on a base of 83 percent—a difference that is statistically significant though modest in magnitude. Appendix **Table I.5** repeats the analysis for Younger Allottees. There the school-attendance difference persists (1.8 percentage points) and is joined by more favorable labor-market conditions: an employment-to-population ratio 2.7 percentage points higher, an unemployment rate 2.4 percentage points lower, and a marginally lower poverty rate. Some sorting into better neighborhoods therefore cannot be ruled out, particularly for Younger Allottees, though the differences that do appear are small relative to the individual-level gains documented above.

Taken together, these patterns suggest that the gains documented above primarily reflect what treated allottees invested in—schooling, occupations, and homes—rather than relocation into substantially better neighborhoods, though the attendance and, for the younger cohort, labor-market differences leave room for modest sorting on those margins.

Appendix I reports two additional exercises probing the robustness of these results. First, restricting the comparison group to allottees whose land received a dry well rather than a producing one isolates oil wealth from selection on land quality or personal acumen; the qualitative pattern of the main results holds in this smaller sample, though several estimates lose precision. Second, re-estimating the main specifications after excluding all hand-linked census matches addresses the possibility that linkage selection drives the results; point estimates retain the same sign and similar magnitude as in the main sample, with some coefficients losing precision in the smaller automated-only sample.

9 Conclusion

The early-1900s transfer of land to nearly seven thousand Black people in Oklahoma, and the subsequent discovery of lucrative oil deposits under a quasi-random subset of those pieces of land, provides a unique natural experiment. This episode gives me the opportunity to assess the role that post-emancipation stock of wealth played in the economic progress of Black Americans, providing insight into what types of post-emancipation policies may have been effective at reducing long-term racial inequalities, had they been successfully implemented in the decades following the abolition of slavery.

This is the first study to causally estimate the long-run effects of a wealth shock on the socioeconomic outcomes of descendants of the enslaved, addressing a major gap in our understanding of racial wealth disparities. I find that receiving a windfall in the decades following emancipation had positive economic effects that persisted over time and even across generations. In the short run, oil wealth led to urbanization: by 1920, treated Creek Freedmen were more likely to live in urban areas, where educational and economic opportunities were concentrated—though these moves were local, keeping treated families largely within the county of their allotment and Creek Nation territory. Over time, treated allottees stayed in school longer and were more likely to enter professional occupations, and their children attended school at higher rates—without any offsetting reduction in family size. These patterns suggest that human capital investments were a key channel through which wealth translated into lasting gains. The intergenerational gains ran through mothers: oil income accruing to women translated most directly into children's schooling. Treated families also appear to have protected the physical wealth they held: despite relocating to cities, where homeownership was far less common, they owned homes at rates comparable to untreated allottees and were less likely to lose a home during the 1920s.

While this study provides valuable causal evidence, further research is needed on longer-term and intergenerational effects, as well as potential policy implications. In ongoing work, I am working to characterize the size, cadence and duration of the wealth shocks. Understanding what types of monetary shocks were most effective in reducing racial disparities in the past is key to designing future policy that is targeted to reduce inequality.

My research demonstrates that access to wealth can be transformative, underscoring both the persistent impacts of historical injustices and the potential for targeted interventions to promote economic mobility.

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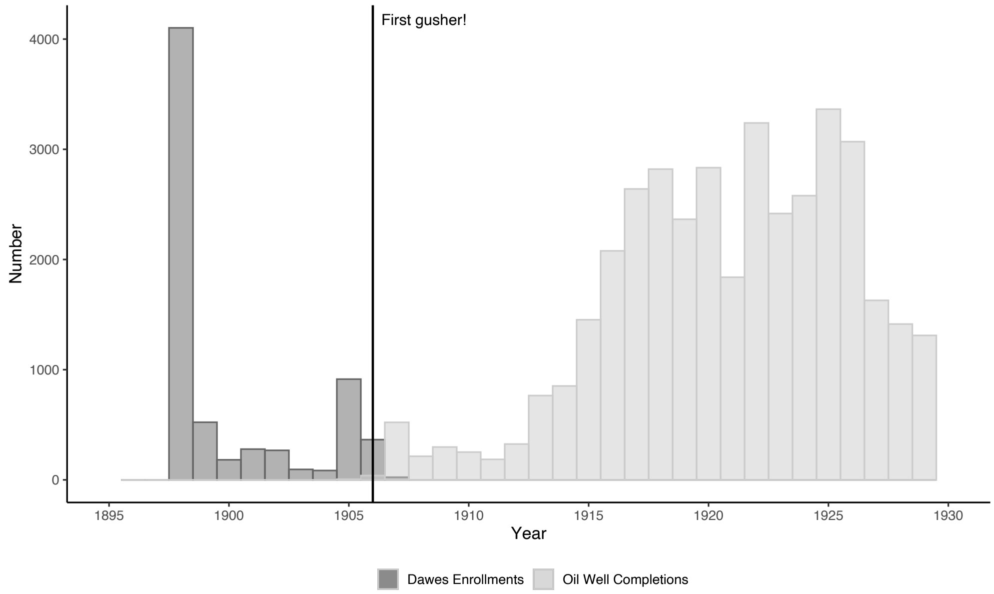
List of Figures

Figure 1: Freedmen Enrollment in the Muscogee (Creek) Nation



Note: This figure plots the year that each Freedmen allottee was enrolled as a citizen in the Muscogee (Creek) Nation by the Dawes Commission.

Figure 2: Muskogee Times-Democrat, September 4, 1913

Little Sarah Will Soon Be In The Plute Class

To have an income larger than that of the president of the United States, or of a life insurance or railroad president, and all out of one oil well, is the good fortune of Sarah Rector, a little ten year old negro girl, living just west of Muskogee near Crekola.

When Sarah's parents applied for an allotment for her, as a descendant of a Creek freedman, she was allotted her portion near what is now the town of Cushing. Lately oil derricks began to spring up about her land, and finally a lease was made on Sarah's allotment.

A well was drilled, and a few days ago, the famous Jones well, the largest ever tapped in the Mid-Continent oil field, was brought in on her land. The well is making 2,500 barrels of oil a day, and has been widely advertised as "the well that produces over $2,500 a day, or over $2 a minute."

Many miles from the property, however, little Sarah continues at her daily play, and plans what she is going to do with the $300 a day which she gets as royalty. Her family is well-fixed, but in a short time little Sarah will be the richest of them all. As other wells are drilled, her income will multiply by leaps and bounds, so that ultimately not even John D. or Vincent Astor will have anything on Sarah in the matter of ready cash for ice cream and such delights of the very young.

GAYNOR OFF FOR TRIP TO EUROPE

New York, Sept. 4.—Mayor Gaynor surprised friends and political enemies alike today by sailing for Europe. Astonishment that he should leave New York on the eve of his campaign for re-election was somewhat allayed by the announcement that his trip would last only twenty days.

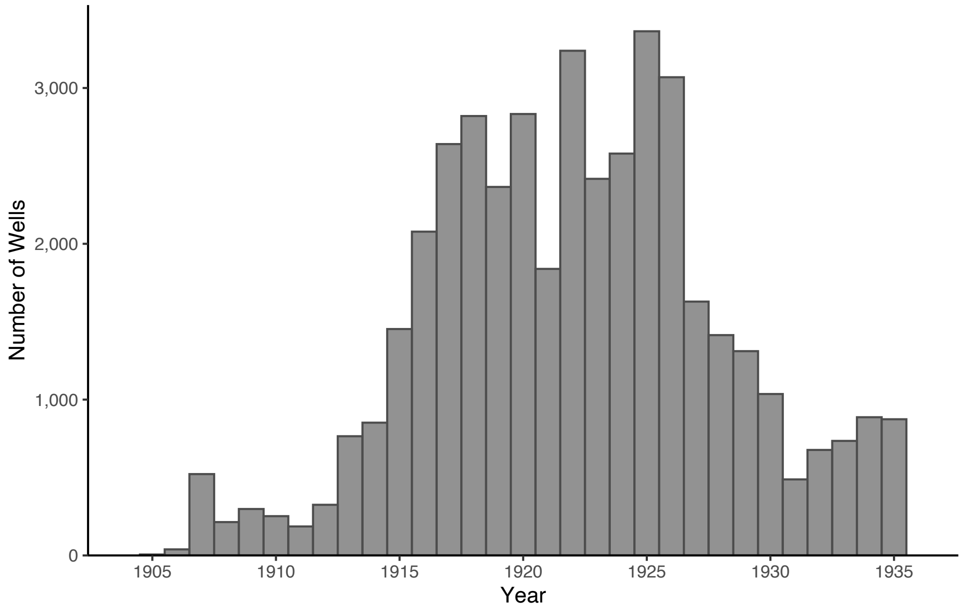
Mayor Gaynor's plan previously announced, spoke of a vacation trip of two weeks to the Adirondacks.

The mayor changed his plans at the eleventh hour because he believed a sea trip would give him greater relief from the throat trouble that has recurred since he was shot more than three years ago. The bullet that lodged in his throat remains there and his voice never has served him for public speaking since then.

There was some doubt today about the composition of the ticket to run with Mayor Gaynor. It has not been announced whether the mayor would accept on his ticket the fusion nominees for comptrollers and president of the board of aldermen.

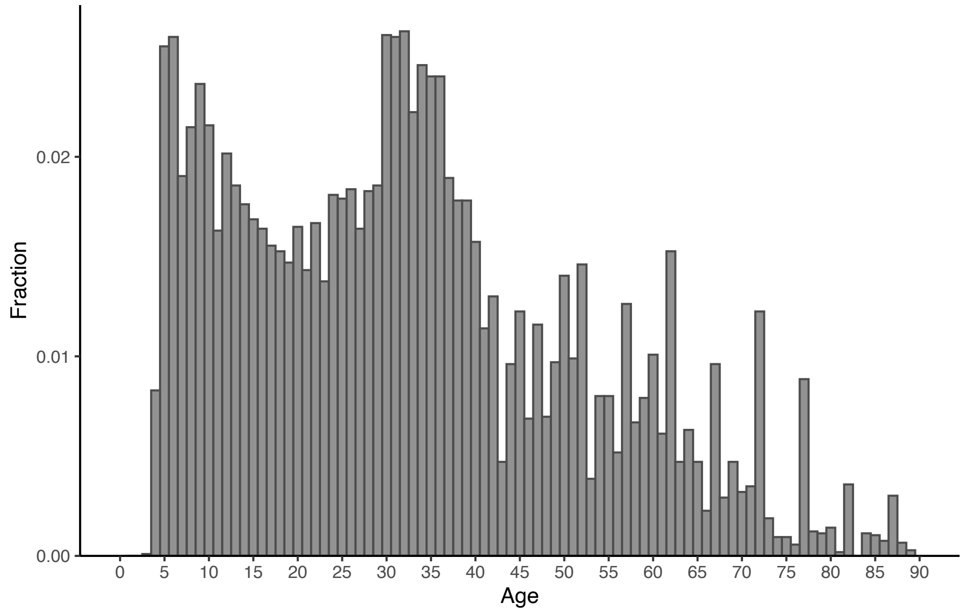
Note: This figure contains a news announcement from the week that the gusher was drilled onto Sarah Rector's allotment. "Plute" is short for plutorat.

Figure 3: Distribution of Drilling Activity within the Muscogee (Creek) Nation



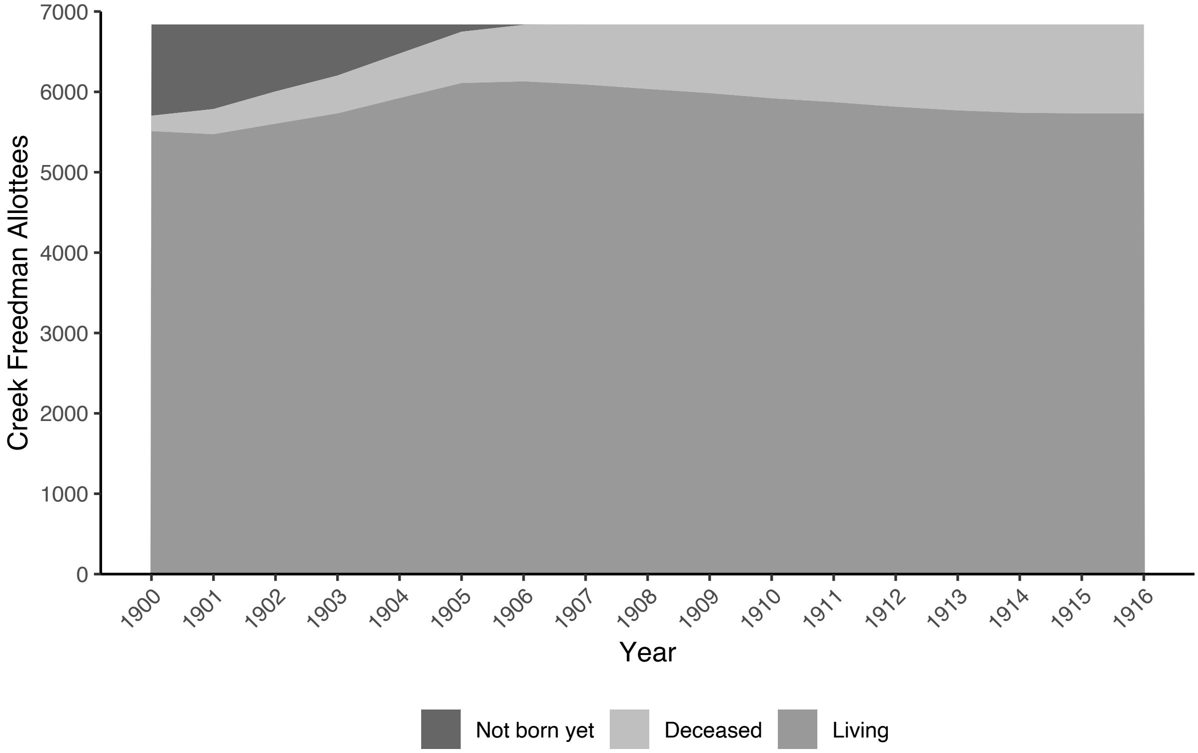
Note: This figure plots the date that each unique well was reported as completed within the Muscogee (Creek) Nation between 1907 and 1935. The Oklahoma Corporation Commission (OCC) was established in 1907 to regulate the Fuel, Oil and Gas, Public Utilities, and Transportation Industries in the new state of Oklahoma. I restrict the wells included in this figure to later than 1907 because some wells drilled prior to OCC oversight may be missing, or their dates of completion may be incorrectly set to 1901. This figure is intended to communicate the timing of oil industry activity in the Muscogee (Creek) Nation and thus includes all categories of wells that oil companies constructed, including oil-producing, dry, gas, injection, and others. For a histogram showing the distribution of "treatment arrival," see **Figure 6**.

Figure 4: Distribution of Creek Freedmen Allottee Ages in 1910



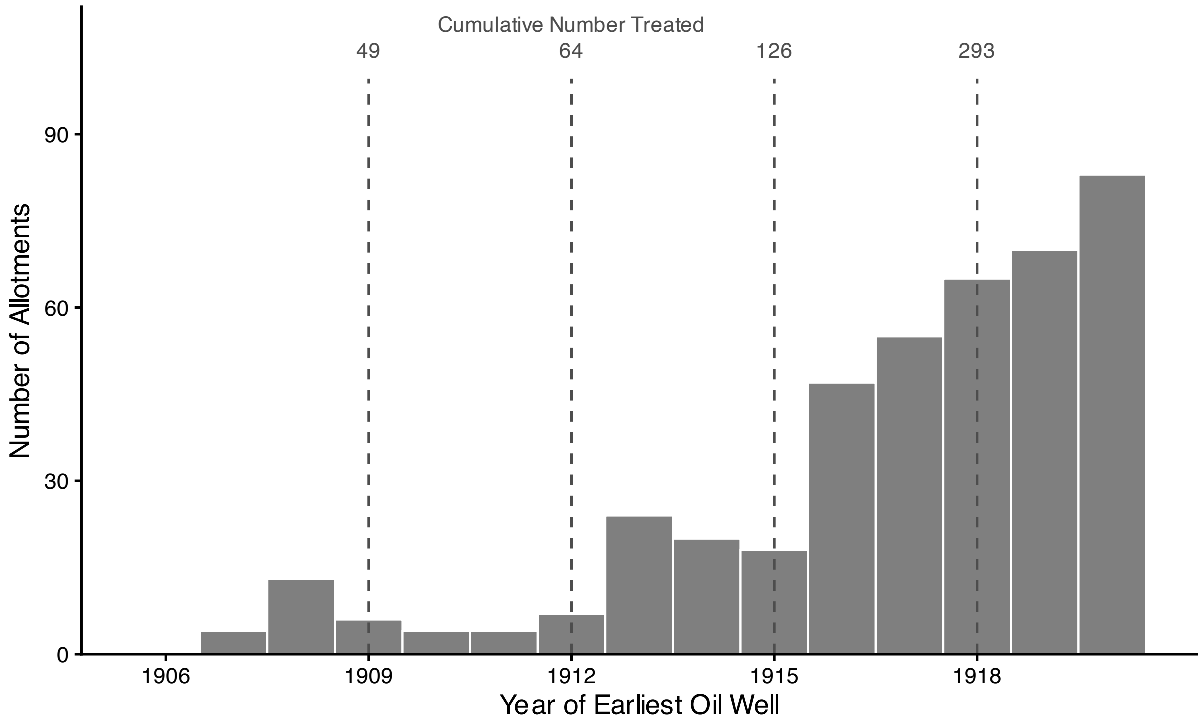
Note: Age is imputed based on "Dawes Census Card Number" and applicant "type." For a discussion on bunching in reported age, see Appendix Subsection H.1. This figure includes all 6,839 Creek Freedmen allottees, regardless of whether they are indicated to have died by 1910. An age histogram set in 1900 would be closer to the time that the Dawes Commission drew up the original Census Rolls and would thus not include quite as many deceased people, but it would exclude the many "Minor" and "New Born" children born between 1900 and 1907.

Figure 5: Living Sample Size Between 1900 and 1915



Note: Freedmen allottees in each group calculated from recorded birth and death dates. Dates are incomplete (see Appendix Section D).

Figure 6: Distribution of Year of First Oil Production Under Each Allotment



Note: This figure plots allotments originally owned by Creek Freedmen on which a producing oil well was drilled by 1920, by the year that the allotment first received a producing oil well. The small number of allotments that first produced oil before 1906 are omitted from the histogram but included in the cumulative counts. The dashed lines occur at three-year intervals (1909, 1912, 1915, and 1918) and report the cumulative number of treated allottees by each year. The 1918 line marks the cutoff used in the treatment definition; drilling in the region continued after 1918 (see **Figure 3**).

List of Tables

Table 1: Treatment Arrival

	Cumulative (year < 1925)		Newly Treated	
Year	No	Yes	No	Yes
1905	6,813	26	6,813	26
1910	6,786	53	6,812	27
1915	6,713	126	6,766	73
1920	6,393	446	6,519	320
1925			6,611	228
1930			6,717	122
1935			6,763	76

Note: This table indicates whether oil was yet produced on allotment land originally owned by the allottee over five-year intervals. The first and second columns sum to 6,839, the total population of Creek Freedmen allottees, in all rows.

Table 2: Human Capital Outcomes for Young Allottees

	Panel A: Enrolled in School by Age						Panel B: Occupational Status		
	Age 12 (1)	Age 14 (2)	Age 16 (3)	Age 18 (4)	Age 20 (5)	Age 22 (6)	White Collar (7)	Blue Collar (8)	Farming (9)
Oil	-	-	0.096*** (0.029)	0.236*** (0.090)	0.259* (0.138)	0.144 (0.097)	0.113*** (0.040)	-0.103* (0.058)	-0.010 (0.055)
Mean	1.00	0.98	0.88	0.57	0.18	0.08	0.06	0.70	0.24
Observations	1,125	957	644	439	362	428	1,586	1,586	1,586

Note: Sample is Creek Freedmen allottees born after 1892, matched to at least one census decade. Panel A: Dependent variables are binary indicators for school enrollment at the indicated age, constructed via monotone inference from IPUMS enrollment records across the 1910–1930 censuses: an allottee observed in school at age X is coded as enrolled at all target ages $\leq X$; one observed not in school at age X is coded as not enrolled at all target ages $\geq X$. Completed years of education reported in the 1940 census extend the inference for individuals whose census observations fall outside their school-age years, assuming school entry at age 6. Panel B: OLS regressions on three binary indicator variables for class of occupation. All specifications control for age in 1910 (linear and quadratic) and sex. Standard errors clustered by family in parentheses. Treatment is defined as the presence of a producing oil well on own allotment by the end of 1918. For more information on data and variable definitions, see Data Appendix H. Dashes mark cells where the outcome does not vary among treated allottees at the indicated age; the linear probability model is uninformative at that boundary, so the coefficient and standard error are suppressed while the mean and sample size are retained. ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$

Table 3: Fertility

	Women (Oil) (1)	Men (Oil) (2)	Pooled (Oil) (3)
Panel A: Offspring count			
By age 40	0.725	-0.486	0.122
	(0.582)	(0.355)	(0.347)
Mean	3.137	2.556	2.810
N	386	498	884
By age 50	1.183	-0.732	0.209
	(0.841)	(0.556)	(0.516)
Mean	3.560	3.326	3.428
N	243	316	559
Panel B: School attendance, ages 6–17			
	0.063**	-0.029	0.051**
	(0.030)	(0.052)	(0.025)
Mean	0.831	0.831	0.831
N	2,234	2,234	2,234

Note: Each coefficient in the fertility panels is from an OLS regression of the outcome on an indicator for whether the allottee was treated as defined in the Data Appendix, controlling for age in 1910, age in 1910 squared; the pooled specification also controls for sex. The Panel A sample is female Creek Freedmen allottees (column (1)), male allottees (column (2)), or both (column (3)), allottees with no linked children are included with an offspring count of zero. Number of offspring by age x counts children observed in census records linked to the allottee. The numbers reflect a substantial effort at deduplication, but instances of individuals observed in different decades but counted as multiple offspring likely remain. Because offspring counts are coded missing when the allottee's last census observation predates the year they would have reached age x, the by-age-40 sample is limited to cohorts born by 1900 (84 percent Older Allottees) and the by-age-50 sample to cohorts born by 1890 (Older Allottees only). Panel B: the sample is children of Creek allottees ages 6–17 observed in the 1920, 1930, or 1940 census. Column (1) ((2)) reports the effect of the mother's (father's) treatment on the child's probability of school attendance; column (3) uses an indicator for treatment of either parent. Panel B controls include child age, child age squared, child sex, and census year fixed effects. Standard errors clustered by Family ID. *** $p < .01$, ** $p < .05$, * $p < .1$

Table 4: The Effect of Oil Money on Labor Force Entry and Household Formation for Allottees

Outcome	Age 16 (1)	Age 18 (2)	Age 20 (3)	Age 22 (4)	Age 24 (5)	Age 26 (6)	Age 28 (7)	Age 30 (8)
Panel A: Men								
In school	-	0.108	0.170	0.156	-	-	-	-
		(0.150)	(0.170)	(0.121)				
Mean	0.90	0.58	0.17	0.09	0.03	0.02	0.01	0.01
N	373	267	221	275	341	481	574	597
In workforce	-0.090	-0.140	-0.164	-0.125	-0.074	-0.044	-0.028	-
	(0.110)	(0.113)	(0.104)	(0.097)	(0.070)	(0.052)	(0.034)	
Mean	0.36	0.67	0.84	0.91	0.94	0.97	0.99	1.00
N	363	368	366	389	434	538	592	609
Ever married	-	-	0.002	0.002	0.283**	0.181*	0.120*	0.065
			(0.086)	(0.126)	(0.123)	(0.095)	(0.064)	(0.054)
Mean	0.01	0.02	0.08	0.19	0.34	0.61	0.79	0.85
N	551	397	286	256	273	327	382	409
Head/spouse	-	-	0.009	-0.040	0.148	0.106	0.123	0.090
			(0.047)	(0.053)	(0.097)	(0.115)	(0.113)	(0.105)
Mean	0.00	0.01	0.04	0.09	0.17	0.32	0.52	0.61
N	642	522	432	395	399	390	390	382
Panel B: Women								
In school	0.077	0.364***	0.408	0.116	-	-	-	-
	(0.063)	(0.088)	(0.244)	(0.174)				
Mean	0.86	0.55	0.18	0.08	0.03	0.01	0.01	0.01
N	271	172	141	153	200	267	297	306
Ever married	-	-0.013	0.049	0.195	0.134	0.180*	-	-
		(0.073)	(0.167)	(0.218)	(0.162)	(0.094)		
Mean	0.01	0.06	0.19	0.32	0.53	0.74	0.88	0.90
N	342	217	139	121	137	168	169	181
Head/spouse	-	-	0.006	0.136	0.083	0.005	0.109	0.057
			(0.088)	(0.143)	(0.145)	(0.158)	(0.147)	(0.136)
Mean	0.01	0.03	0.08	0.13	0.28	0.45	0.63	0.70
N	405	282	202	172	168	161	144	144

Note: Sample is Creek Freedmen allottees born after 1892, matched to at least one census decade. Dependent variables are binary indicators for the outcome listed, measured at the indicated age via monotone inference across the 1910–1940 censuses. Specifications are estimated separately by gender and control for age in 1910 (linear and quadratic). Standard errors clustered by family in parentheses. Treatment is defined as the presence of a producing oil well on own allotment by the end of 1918. For more information on data and variable definitions, see Data Appendix H. Dashes mark cells where the outcome does not vary among treated allottees at the indicated age; the linear probability model is uninformative at that boundary, so the coefficient and standard error are suppressed while the mean and sample size are retained. $***p < 0.01$, $**p < 0.05$, $*p < 0.1$

Table 5: The Effect of Oil Money on Occupational Status: Multinomial Logit Results

	Older Allottees		Younger Allottees		Pooled	
	White Collar (1)	Farming (2)	White Collar (3)	Farming (4)	White Collar (5)	Farming (6)
Panel A: Log Odds						
Oil	0.431	-0.136	1.281***	0.121	0.962***	-0.019
	(0.508)	(0.258)	(0.325)	(0.309)	(0.301)	(0.208)
Panel B: Average Marginal Effects						
Oil	0.022	-0.042	0.114***	-0.009	0.065**	-0.029
	(0.026)	(0.058)	(0.041)	(0.052)	(0.028)	(0.041)
N	1578	1578	1586	1586	3164	3164

Note: Each column presents a multinomial logit equation for the named occupational outcome relative to blue-collar employment, which serves as the base category. Panel A: Reported coefficients are log-odds ratios. Panel B: average marginal effects. Treatment is defined as the presence of a producing oil well on the allottee's own allotment by the end of 1918. All specifications include year fixed effects and control for age in 1910, age in 1910 squared, and an indicator for male. Standard errors clustered at the Family ID level are reported in parentheses. *$p < 0.10$, **$p < 0.05$, ***$p < 0.01$.

Table 6: The Effect of Oil Money on Urbanization

	Older Allottees				Younger Allottees				Pooled			
	Urban	Stayer	Urban(S)	Urban(M)	Urban	Stayer	Urban(S)	Urban(M)	Urban	Stayer	Urban(S)	Urban(M)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Oil	0.078	-0.184***	-0.039	0.161*	0.195**	-0.107**	0.140	0.194**	0.137***	-0.151***	0.031	0.179**
	(0.058)	(0.041)	(0.050)	(0.089)	(0.080)	(0.050)	(0.120)	(0.098)	(0.053)	(0.033)	(0.058)	(0.073)
Mean(Y)	0.193	0.719	0.111	0.266	0.242	0.618	0.144	0.331	0.218	0.674	0.128	0.300
N	965	3,837	458	507	1,050	2,997	500	550	2,015	6,834	958	1,057

Note: This table presents regression estimates of the effect of oil wealth on urbanization and related outcomes. Outcomes include: living in an urban area in 1920, remaining in one's 1910 census township through 1920 (Stayer), 1920 urban residence conditional on remaining (Urban(S)), and 1920 urban residence conditional on moving (Urban(M)). The sample comprises allottees matched in both the 1910 and 1920 censuses. Specifications are estimated separately for Older Allottees and Younger Allottees, and pooled. Treatment is defined as the presence of a producing oil well on own allotment by the end of 1918. Standard errors clustered at the Family ID level are shown in parentheses. ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$

Table 7: The Effect of Oil Money on Homeownership

	Older Allottees		Younger Allottees	
	Homeowner (1)	House Value (2)	Homeowner (3)	House Value (4)
Oil	0.024	742.676	0.066	2221.924
	(0.057)	(1036.914)	(0.046)	(1618.013)
Mean	0.51	1137.63	0.19	1751.08
N	1958	259	1824	185

Note: This table presents regression estimates of the 1920–1940 pooled differences in homeownership between treated and untreated Creek Freedmen allottees who were born before or after 1892. All regressions control for age, age squared, and gender, and the pooled regression controls for year. Means are computed over the full treated and untreated Creek Freedmen allottee population. Treatment is defined as the presence of a producing oil well on own allotment by the end of 1918. For more information on data and variable definitions, see Data Appendix H. Standard errors are clustered by family groups and are shown in parentheses. ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$

Table 8: Mediator Analysis: Oil → Urbanization → Homeownership

	Older Allottees			Younger Allottees		
	Urban	Homeowner (Total)	Homeowner (Direct)	Urban	Homeowner (Total)	Homeowner (Direct)
	(1)	(2)	(3)	(4)	(5)	(6)
Oil	-0.001	0.024	0.023	0.135**	0.066	0.075
	(0.048)	(0.057)	(0.057)	(0.067)	(0.046)	(0.046)
Urban			-0.115***			-0.066***
			(0.031)			(0.021)
Mean	0.24	0.51	0.51	0.40	0.19	0.19
N	1958	1958	1958	1824	1824	1824

Note: This table presents regression estimates of the effect of oil wealth on homeownership via urbanization as a potential mediator. Columns report: oil's total effect on urban residence and homeownership (Columns 1–2); oil's direct effect on homeownership conditional on urban status (Column 3), which removes the indirect effect operating through urbanization; and the effect of urban residence on homeownership (Column 4), which quantifies the mediator relationship. All regressions pool the 1920–1940 censuses with year fixed effects; specifications are estimated separately for Older Allottees and Younger Allottees. Treatment is defined as the presence of a producing oil well on own allotment by the end of 1918. Standard errors are clustered at the Family ID level and shown in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$

Table 9: The Effect of Oil Money on Migration

	(1)	(2)
Panel A: Left County		
Coefficient	-0.077*	-0.058
Std. error	(0.044)	(0.040)
Mean	0.67	0.67
Observations	4,261	4,261
Panel B: Left Creek Nation		
Coefficient	-0.063*	-0.038
Std. error	(0.032)	(0.032)
Mean	0.25	0.25
Observations	4,261	4,261
Panel C: Left Oklahoma		
Coefficient	0.010	0.005
Std. error	(0.029)	(0.030)
Mean	0.14	0.14
Observations	4,261	4,261
County FE?	No	Yes

Note: This table presents regression estimates of the difference in likelihood of being observed out of Oklahoma, out of the Muscogee (Creek) Nation, or out of allotment county, respectively, between treated and untreated Creek Freedmen allottees in 1920, 1930, and 1940. I define allotment county as the Oklahoma county in which the allottee's Dawes Enrollment was located. Regressions control for age, age squared, and gender and include fixed effects for census decade. Means are computed over the full treated and untreated Creek Freedmen allottee population. Treatment is defined as the presence of a producing oil well on own allotment by the end of 1918. For more information on data and variable definitions, see Data Appendix H. Standard errors are clustered at the Family ID level and are shown in parentheses. ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$

Table 10: Neighborhood Quality, 1940

Outcome	Estimate	Mean	N
Panel A: Education			
Share graduated college	0.037	0.14	1,003
	(0.039)		
Share graduated high school	0.028	0.28	1,003
	(0.034)		
School attendance	0.015**	0.83	1,003
	(0.006)		
Teacher Salary	62.342	937.70	935
	(44.865)		
Panel B: Labor Markets			
Employment to population ratio	-0.007	0.81	1,003
	(0.016)		
Unemployment rate	0.008	0.13	1,002
	(0.014)		
Racial earnings gap	-26.577	490.74	821
	(43.433)		
Panel C: Families			
Share of families single-parent	0.011	0.17	1,001
	(0.013)		
Poverty rate	-0.001	0.70	1,003
	(0.022)		
Panel D: Segregation			
Proportion of neighbors white	-0.074*	0.58	1,003
	(0.043)		
Panel E: Other			
Urban	0.000	0.44	1,007
	(0.058)		

Note: This table presents regression estimates of selected neighborhood characteristics of the places where treated and untreated Creek Freedmen allottees lived in 1940. All regressions control for age, age squared, and gender. Means are computed over the full treated and untreated Creek Freedmen allottee population. Treatment is defined as the presence of a producing oil well by the end of 1918 on own allotment. For more information on data and variable definitions, see Data Appendix H. Standard errors are clustered over Family ID and are shown in parentheses. ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$

A Additional Outcomes

B Additional Descriptive Statistics

C News Items

D Census Linking

For each match decade, I first subset my Creek Freedmen database to people who are likely to have been living in the relevant year, although this information is by no means complete.[21] Importantly, there is considerable bunching within reported age for small children in particular. See **Figure I.9** and **Figure I.10**.

- **Cleaned Name.** I standardize names via the *abeclean* Stata command which converts nicknames into likely given names by gender, such that "Sam" would become "Samuel" for a man, and "Samantha" for a woman (Curtis and Eriksson, 2020). I also output NYSIIS names.
- **Enumeration District, 1900 (census).** To assist in linking, I manually classified all post-office locations into 1900 Census enumeration districts (EDs).

I link Creek Freedmen allottees to the 1910-1920 Censuses separately and iteratively, using a combination of automated and manual methods.[22]

1. In Stage 1, I employ a novel household-linking strategy.[23] Taking the other members of the household into account allows me to approve, with near certainty, many candidates in whom I would not otherwise have felt confident.
    (a) Using the Stata commands *matchit* and *reclink2*, I search broadly for each individual by standardized and unstandardized first and last name over all Black and Native American people living in Indian Territory within a cell defined only by sex. I keep those matches within a generous band around reported age such that each allottee has many candidates.
    (b) I identify *census households* for which multiple individuals from the same *household in the Creek Freedmen database* match.[24] I reshape the linked file to be at the household level, rather than the individual level.
    (c) I review household-level candidates by hand using the Stata command *clrevmatch*.
    (d) I subset the expansive data from 1a to only those individual candidates from matched households identified in 1c. I check the matched dataset for duplicates over *Roll_Number* + *Type* and over *histid*.
2. In Stage 2, I take advantage of location data that was collected by the Dawes Commission: the nearest post office to where the application took place.
    (a) I start with the expansive data from 1a and drop all candidates identified by *Roll_Number* + *Type* and *histid* for whom matches have already been identified.
    (b) Using the Stata command *abematch*, I search for each individual by age and unedited first and

[21] My information on deaths is incomplete and in particular, would only include deaths by 1914 at the latest.

[22] Indian Territory was enumerated in the 1900 Census but is unavailable on IPUMs.

[23] My strategy differs from Helgertz et al. (2022).

[24] I define a "household" in the Creek Freedmen database as individuals listed on the same Dawes Census Card, inclusive of parents listed on different cards than their children.

last name among individuals in that ED.[25]

   (c) I return the remainder of matches to the common pool.

3. I next search for each individual by age and standardized first and last name among individuals in that enumeration district.
   (a) I append the near certain matches from Stage 1 to this.
   (b) I identify *census households* for which multiple individuals from the same *household in the Creek Freedmen database* match, now taking these as near-certain.
4. I generate a set of more ambiguous matches to be reviewed by hand by a group of research assistants and myself.
5. **Automated Links**: In the short term, I supplement my hand links with automated links. Using the Stata command *abematch*, I search for each individual by sex, age, race and NYSIIS-standardized first and last name among individuals born in Oklahoma. I allow for age to be up to five years incorrect on either end but matches with smaller errors will take priority over those with larger errors. In the standard version, which I utilize here, this command discards Freedmen that match to multiple individuals in the Census, thereby preserving only Freedmen who match to a single individual in the Census. I keep matches for which variable output "uniquefile1" is equal to 1, meaning that, IN EACH FILE, individuals are unique by NYSIIS name within a year above and below their reported age.

[25]Because *abematch* only identifies unique matches, there can be large sample size gains from matching within covariate groups defined as precisely as possible, as long as the groups are credible.

E Laws of Descent and Distribution

E.1 Restrictions on the Sale of Allotments

Original

Sec. 7. "Lands allotted to citizens hereunder shall not in any manner whatsoever or at any time be incumbered taken or sold to secure or satisfy any debt or obligation contracted or incurred prior to the date of the deed to allottee therefor and such lands shall not be alienable by the allottee or his heirs at any time before the expiration of five years from the ratification of this agreement, except with the approval of the Secretary of the Interior.

Each citizen shall select from his allotment forty acres of land as a homestead which shall be nontaxable and inalienable and free from any incumbrance whatever for twenty-one years, for which he shall have a separate deed, conditioned as above: *Provided,* That selections for minors, prisoners, convicts, incompetents, aged and infirm persons who can not select for them may be made in the manner herein provided selection of their allotments; and if for any reason selection be not made for any citizen it shall be of said commission to make selection for him.

The homestead of each citizen shall remain, after the death of the allottee, for the use and support of children born to him after the ratification of this agreement but if he have no such issue then he may dispose of his homestead by will free from limitation herein imposed, and if this be not done, the land shall descend to his heirs according to the laws of descent and distribution of Muscogee (Creek) Nation, free from such limitation."

Original Creek Agreement (31 Stat. L. 861) as given in Kappler (1904), page 730.

First Removal of Restrictions

"And all the restrictions upon the alienation of all land of allottees of either of the Five Civilized Tribes of Indians who are not of Indian blood, except minors, and except as to homesteads, hereby removed, and all restrictions upon the alienation of all other allottees of said tribes, except minors, and except as to homesteads, may, with the approval of the Secretary of the Interior, be removed under such rules and regulations as the Secretary of the Interior may prescribe, upon application to the United States Indian agent at the Union Agency in charge of the Five Civilized Tribes, if said agent is satisfied upon a full investigation of each individual case that such removal of restrictions is for the best interest of said allottee."

Act of April 21, 1904 (33 Stat. 189.) as given in Kappler (1913), page 50.

Addition of Restrictions (Full-Blood Indian)

Sec. 19. "That no full-blood Indian of the Five Tribes shall have the power to alienate, sell, dispose of, or encumber in any manner any of the lands allotted to him for a period of 25 years from and after the passage and approval of this act..."

Act of April 26, 1906 (34 Stat. L. 144) as given in Kappler (1913), page 176.

Sale Upon Death of Allottee

Sec. 22. "That the adult heirs of any deceased Indian of either of the Five Civilized Tribes... may sell and convey the lands inherited from such a decadent; and if there be both adult and minor heirs of such decadent, then such minors may join in a sale of such lands by a guardian..."

Act of April 26, 1906 (34 Stat. L. 137) as given in Kappler (1913), page 178.

Second Removal of Restrictions

"All lands, including homesteads, of said allottees enrolled as intermarried whites, as freedmen, and as mixed-blood Indians having less than half Indian blood including minors shall be free from all restrictions."

Act of May 27, 1908 (35 Stat. 312.) as given in Kappler (1913), page 351.

E.2 Birthdate of Inclusion on the Dawes Rolls

"Citizen" Rolls

Sec. 28. "No person, except as herein provided shall be added to the roles of citizenship of said tribe after the date of this agreement and no person whomsoever shall be added to said rolls after the ratification of this agreement.

All citizens who were living on the first day of April, eighteen hundred and ninety-nine, entitled to be enrolled under section twenty-one of the Act of Congress approved June twenty-eight, eighteen hundred and ninety-eight, entitled "An Act for the protection of the people of Indian Territory, and for other purposes," shall be placed upon the rolls to be made by said commission under said Act of Congress, and if any such citizen has died since that time, or may hereafter die, before receiving his allotment of lands and distributive share of all the funds of the tribe, the lands and money to which he would be entitled, if living, shall descend to his heirs, according to the law of descent and distribution of the Muscogee (Creek) Nation, and be allotted and distributed to them accordingly.

All children born to citizens so entitled to enrollment, up to and including the first day of July, nineteen hundred, and then living, shall be placed on the rolls made by said commission; and if any such child die after said date, the lands and moneys to which it would be entitled, if living shall descend to its heirs according to the laws of descent and distribution of the Muscogee (Creek) Nation, and be allotted and distributed to them accordingly."

Original Creek Agreement (31 Stat. L. 861) as given in Kappler (1904), page 736.

New Born Rolls

"That the commission to the Five Civilized Tribes is authorized for sixty days after the date of approval of this act to receive and consider applications for enrollment of children born subsequent to May twenty-five, nineteen hundred and one, and prior to March fourth, nineteen hundred and five, and living on said latter date, to citizens of the Creek Tribe of Indians whose enrollment has been approved by the Secretary of the Interior prior to the date of the approval of this act; and to enroll and make allotments to such children."

Act of March 3, 1905 (33 Stat. L. 1048) as given in Kappler (1913), page 148.

Minor Rolls

Sec. 2. "That for ninety days after approval hereof applications shall be received for enrollment of children who were minors living March fourth, nineteen hundred and six, whose parents have been enrolled as members of the [Five Civilized Tribes], or have applications for enrollment pending at the approval hereof, and for the purposes of enrollment under this section illegitimate children shall take the status of the mother, and allotments shall be made to children so enrolled."

Act of April 26, 1906 (34 Stat. L. 137) as given in Kappler (1913), page 170.

Determination of Race and Age of Majority

"That the rolls of citizenship and of freedmen of the Five Civilized Tribes approved by the Secretary of the Interior shall be conclusive evidence as to the quantum of Indian blood of any enrolled citizen or freedman of said tribes... and the enrollment record... hereafter be conclusive evidence as to the age of said citizen or freedman."

Act of May 27, 1908 (35 Stat. 312.) as given in Kappler (1913), page 352.

E.3 Heirship

District Courts and Arkansas Law

"That there shall be appointed... four additional judges of the United States court in the Indian Territory... And said judges shall have all the authority and exercise all the powers, perform like duties, and receive the same salary as other judges of said court...

Sec. 2. All the laws of Arkansas heretofore put in force in the Indian Territory are hereby continued and extended in their operation, so as to embrace all persons and estates, whether Indian, freedman or otherwise, and full and complete jurisdiction is hereby conferred upon the district courts in said Territory in the settlements of all estates of decadents, the guardianships of minors and incompetents, whether Indians, freedmen or otherwise."

Act of April 28, 1904 (33 Stat. 573.) as given in Kappler (1913), page 109.

Wills

Sec. 23. "Every person of lawful age and sound mind may by last will and testament devise and bequeath all of his estate, real and personal, and all interest therein."

Act of April 26, 1906 (34 Stat. L. 137) as given in Kappler (1913), page 178.

E.4 Taxes

"That all the lands upon which restrictions are removed shall be subject to taxation, and the other lands shall be exempt from taxation as long as the title remains in the original allottee."

Act of April 26, 1906 (34 Stat. L. 137) as given in Kappler (1913), page 177.

F Oil Leases

1. Oil companies hire "lease men" to make "advanced royalty" leases with small landholders for right to dig exploratory holes in their land
   (a) Lease man identifies land that they might like to bet on.
   (b) Lease man contacts the landholder and leases land for a signing bonus, plus \$24-48 annually (\$830-1600) regardless of the drilling status of the land.
2. If land is "restricted Indian" land, the lease man must follow terms of the Bureau of Indian Affairs (BIA) and must seek approval of the BIA before it goes into effect. While the exact regulations changed with time, they included stipulations like minimum royalty shares (i.e. 12.5%) and maximum lengths of time that the leaseholder was allowed to hold lease without drilling. A sample of a notice of oil lease regulation changes, made by the Acting Commissioner of the Indian Affairs at the Department of Interior, dated June 1907, can be found at **Figure I.4**.
3. Within next few months, driller drills hole(s)
4. If well "produces" oil, landholder gets stipulated royalty share of revenue.

G Maps

H Data Appendix

H.1 Allottees: Dawes Census Cards

I digitized *Campbell's Abstract of Creek Freedman Census Cards and Index*, a typed and published volume of every Creek Freedman Dawes Census Card (Campbell, 1915a).[26] These cards provide a small number of variables:

- **Post Office.** Allottees usually applied for Creek enrollment with the traveling Bureau of Indian Affairs (BIA). The post office nearest to where the enrollment took place is generally listed as the location.
- **Dawes Roll Number.** Each allottee was assigned a persistent "Dawes roll number" or just "roll number," unique within his or her enrollment category: Creek Indian, Freedmen, Minor, or New Born. Minor and New Born are additionally split into Creek Indian or Freedman, for a total of six enrollment categories within the Muscogee (Creek) Nation.
- **Name.**
- **Age.** I assume that the age listed on the Dawes Census Cards is the age reported at the date of enrollment and not, for example, age at the date of application.[27] There was considerable bunching within reported age, particularly for infants (**Figure I.9** and **Figure I.10**). If we assume that the census links are correct, then it seems that many of the parents reported their children to have been born earlier to the Dawes Commission than they did to the census enumerators. This behavior may have been strategic, in response to the three successive enrollment birthdate cutoffs.
- **Age Group.** I define allottees born after 1892 and born prior to March 4, 1906 as Younger Allottees, and allottees born earlier as Older Allottees. Younger Allottees include all enrollees on the Minor and New Born rolls as well as the youngest allottees on the main Creek Freedmen Rolls. Subsection 2.2 describes the enrollment process and Subsection E.2 gives the full text of the laws.
- **Sex.**

[26] It is fortunate that this typed and published volume exists, as the Dawes Rolls for some nations still only exist as handwritten cards.

[27] Unfortunately, even contemporary experts were not entirely confident in this interpretation. On page 4, Campbell (1915a) writes that "many believe that the date of enrollment is the date from which you calculate to determine the age of the allottee. In a great many instances this is true, but in many cases this is not true. The law says, that the enrollment record shall govern as to age. The enrollment record may include the testimony taken at the time of enrollment, affidavits, birth certificates, etc. Everyone ought before purchasing, or leasing lands, to secure from the Commissioner a certified copy of the enrollment testimony, if there be any question as to age."

H.2 Fertility/School Attendance Sample

I begin with the intersection of the Dawes Rolls and each of the 1910, 1920, and 1930 decennial censuses, identifying enrolled Creek Freedmen observed in each census year. Within each census year, I extract all non-enrolled co-resident household members born before 1925 — individuals with no Dawes Roll Number — for whom at least one directly related household member carries a Dawes Roll Number. Specifically, I require that the individual's mother (`momloc`), father (`poploc`), or at least one co-resident sibling be an enrolled Creek Freedman.[28] This yields the treatment-eligible sample: individuals born into Creek Freedmen families who are not themselves enrolled on the Dawes Rolls — and thus hold no formal rights to Creek land allotments or oil royalties — but whose parents or siblings do.

H.3 Construction of Family Identifiers

Several treatment variables, like oil income received by a parent, sibling, or child, are defined such that many members of the same family share the same treatment value. Within-family outcome correlation arising from shared resources and common household shocks compounds this. Both sources of within-cluster dependence make family-level clustering the appropriate level for inference. I assign each enrolled Creek member a Family ID (`family_id`) using the parent–child linkages recorded on the Dawes Roll. The construction proceeds in two stages: generation classification and family formation.

- **Generation classification.** Each enrolled member is classified as G0 (grandparent), G1 (parent), or G2 (child) based on the mother and father listed on their Dawes Census Card. A person is designated G1 if they appear as someone else's enrolled parent; G2 if their own enrolled parent appears on the roll; and G0 if they are an enrolled parent whose own parents are not on the roll. Individuals who are both an enrolled parent and have an enrolled parent themselves are classified as G1 (they are the generational bridge). 521 members with no roll-based linkage ("Unlinked") are resolved by secondary rules: children under 18 at enrollment are placed into an existing family if their Dawes Census Card shares a G1 or G0 anchor. Remaining unlinked children, often orphans, are grouped into card-only sibling families. Unplaced adults are assigned G0 or G1 by birth before or after 1870. After these steps, every enrolled member receives a generation assignment.
- **Family formation.** Families are anchored by the G1 mother where possible. A mom-family collects all G2 children sharing the same enrolled mother; a dad family collects G2 children whose mother is not enrolled; and card-families capture sibling groups with no enrolled parent. In total, 1,584 families are formed (1,178 mother-anchored, 330 father-anchored, 76 card-only).
- **Person-level Family ID assignment.** Most members belong to exactly one family and receive that

[28] Siblings are identified by shared `momloc` when non-zero, falling back to shared `poploc` for the small number of households where only a father is coresident ($N = 115$ individuals across 33 households in 1910). The IPUMS pointer variables `momloc` and `poploc` give the `pernum` of each individual's mother and father within the household.

family_id directly. Two edge cases arise: G0 grandparents who appear in multiple families (as parents of multiple enrolled G1 children) receive a missing family_id; their generation is still assigned but they are excluded from family-level clustering. G1 fathers who fathered children with multiple enrolled mothers are assigned to the family containing their most recently born G2 children, reflecting the household most likely to have been intact when oil wealth arrived. 248 individuals appear as both G0 in one family and G1 in another; to assign family_id, they are treated as G1 (parent), prioritizing the family in which they hold the parental role.

H.4 Allotments: Plat Maps

Edward Hastain was a member of the Dawes Commission. He drew up and published maps of the full allotted Muscogee (Creek) Nation in Hastain (1910). There is one map per "township," or six-mile by six-mile area of land, for a total of 149 maps. Trained undergraduate research assistants converted all maps, such as the one in **Figure I.11**, into Public Land Survey System (PLSS) data. Each allotment can be coded into a unique polygon within the United States. Each plot of land on each map is labeled with the owner's name and his or her unique Dawes roll number. Thus, this dataset contains the variables:

- **Dawes Roll Number.** Described in Subsection H.1
- **Name.**
- **Public Land Survey System (PLSS) Code.**

H.5 Oil Well Data

By 1905, as the Oklahoma oil boom exploded, in their fervor to discover oil first and to stake claim, independent drillers and nascent oil companies drilled in an uncontrolled manner. There was a growing sense that regulation and oversight was needed to temper the "external damage caused by the uncontrolled production of crude oil overflowing into fields and streams and certain apparent waste of natural gas" (Clark, 1977, p.376). And thus in 1907, during the first year of Oklahoma's statehood, the Oklahoma Corporation Commission (OCC) was established to regulate public utilities. My primary source of oil well data is the "W27 Well Completion List" from the OCC, which has an entry for all well actions taken since 1907, what the action was, the date and the location in both Public Land Survey System (PLSS) format and latitude/longitude. Most of the well actions are not construction of a producing oil well and most take place after my time period of study, into the present day. For this paper, I subset the well actions to construction of producing oil wells that were completed before the early 1930s.

- **Public Land Survey System (PLSS) Code.**

H.6 Treatment

Primary Treatment

Only enrolled Creek Freedmen have values for these variables.

- **Treated (Self).** Variable has a value of 1, treated, if a producing oil well was built on top of the individual's Dawes allotment by the end of 1918, and a value of 0 if one was built later or never built.
- **Spouse Treated.** Variable has a value of 1, treated, for individuals whose spouses, to whom they were married at the time of enrollment, were treated. Only enrolled Creek Freedmen *in the older generation* have values for these variables.
- **Child Treated.** Variable has a value of 1, treated, for individuals whose child, listed as such on the Dawes Roll, was treated. Only enrolled Creek Freedmen *in the older generation* have values for these variables.
- **Sibling Treated.** Variable has a value of 1, treated, for individuals whose parent, listed as such on the Dawes Roll, was treated *by another child*. Only enrolled Creek Freedmen *in the younger generation* have values for these variables.
- **Parent Treated.** Variable has a value of 1, treated, for individuals whose parent, listed as such on the Dawes Roll, was treated. Only enrolled Creek Freedmen *in the younger generation* have values for these variables.

Secondary Treatment

Children, unenrolled offspring or siblings of enrolled Creek Freedmen, can have values for these variables. Children are defined further in Subsection H.2.

- **Father Treated.** Variable has a value of 1, treated, if a producing oil well was built on the individual's father's Dawes allotment by the end of 1918, and a value of 0 if one was built later or never built. Only children whose father is identified as an enrolled Creek Freedman via `poploc` in the 1910, 1920, or 1930 census have values for this variable.
- **Mother Treated.** Variable has a value of 1, treated, if a producing oil well was built on the individual's mother's Dawes allotment by the end of 1918, and a value of 0 if one was built later or never built. Only children whose mother is identified as an enrolled Creek Freedman via `momloc` in the 1910, 1920, or 1930 census have values for this variable.
- **Sibling Treated.** Variable has a value of 1, treated, if a producing oil well was built on the allotment of at least one of the individual's co-resident enrolled siblings by the end of 1918, and a value of 0 if one was built later or never built on any such allotment. Only children with at least one co-resident enrolled Creek Freedman sibling — identified by shared `momloc`, or shared `poploc` when `momloc` is zero— have values for this variable.

Children are assigned the maximum treatment status across all enrolled family members observed during the first 18 years of their life, when they may have been living at home.

H.7 Individual-Level Outcome Variables

Most of my individual-level outcome variables are derived from cleaned and harmonized census records provided by IPUMS (Ruggles et al., 2024a,b).

Occupational Status

Using "occ1950" in IPUMs extracts, I define occupational classes consistent with the codes delineated in Durlauf et al. (2024), Table B.1. Wives of household heads are given husbands' value for these variables.

- **White-Collar, 1910-1940 (census).** Professionals, Semi-professionals, Proprietors, Managers, Officials, Clerical and Sales.
- **Blue-Collar, 1910-1940 (census).** Craftmen, Government Services, Semi-Skilled and Unskilled.
- **Farming, 1910-1940 (census).** Farmers: Owners, Tenants and Managers. I follow Durlauf et al. (2024) and classify Farm Laborers as Unskilled or Semiskilled and therefore Blue-Collar.
- **Unemployed in labor force, 1940 (census)**.
- **Employment wages/salary, 1939 (1940 census).** Respondent's total pre-tax wage and salary income– that is, money received as an employee– in 1939 ("incwage" in IPUMS extract).
- **Self-Employed, 1910-1940 (census).** An individual has a value of 1 for this variable if they report that they work as an employer or on own account, i.e. not for wages or salary ("classwkr" in IPUMS extract).

Wealth and Assets

- **Homeowner, 1910-1940 (census).** An individual has a value of 1 for this variable if they report owning the home that they live in ("ownershp" equal to 10) <u>and</u> relation to household head ("relate") is given as self (101) or spouse (201). All others, including renting heads/spouses or non-heads/spouses in owned homes, receive a value of 0 for this variable.
- **House Value, 1930-1940 (census).** Value of home in contemporary dollars, conditional on household home ownership. In 1930, house value was collected only for *non-farm* owner-occupied units and includes the value of the land. In contrast, in 1940, house value was collected for *all* owner-occupied units, including single-family farmhouses, but excludes the value of the land.
- **Has Live-In Domestic Servant, 1910-1940 (census).** Definition adapted from Kalsi and Ward (2025). Unless the household head is a clergyman ("occ1950" equal to 9) or the operator of a boarding house (752), members of a main family connected to a household head get a 1 for this variable if the household contains an individual fitting one of the following criteria:

  - Listed as one of the following relations to household head ("relate"): housekeeper (1212), maid (1213), cook (1214), nurse (1215), or other probable domestic employee (1216).
  - Listed as a servant ("relate" equal to 1211) or domestic employee (1251) AND give their occupation as teacher ("occ1950" equal to 93), private dressmaker (633), taxi cab driver/chauffeur (682), private household service worker (700-720), attendant (731), charwomen and cleaner (753), cook (754), practical nurse (781) waiter/waitress (784) or gardener (930).
  - Listed as a servant ("relate" equal to 1211) or domestic employee (1251) AND give their industry as private household ("ind1950" equal to 826).
- **Has Live-In Non-Domestic Employee, 1910-1940 (census).** Members of a main family connected to a household head get a 1 for this variable if the household contains an individual fitting one of the following criteria:
  - Listed as other employee to household head ("relate" equal to 1217).
  - Listed as a servant ("relate" equal to 1211) or non-domestic employee (1252) AND give their occupation as farmer ("occ1950" equal to 100, 123), boarding and lodging housekeeper (764), elevator operator (761), porter (780), farm laborer (810-840), or laborer (970).
  - Listed as a servant ("relate" equal to 1211) or non-domestic employee (1252) AND give their industry as agriculture ("ind1950" equal to 105), hotels and lodging places ("ind1950" equal to 836).

Education

- **Literacy, 1910-1940 (census).** Individual 10 years or older can read and write in any language ("lit" in IPUMS extract). This question was not asked during the 1940 Census and thus I define all individuals who report having completed second grade or above as literate ("educd" in IPUMS extract).
- **"Young" literacy, 1910-1940 (census).** Literacy as defined above, but restricted only to individuals whose literacy was not asked during the last census enumeration (i.e. individuals between the ages of 10 and 20).
- **School attendance, 1910-1940 (census)** Individual between the ages of 5 and 17 who attended school or college any time between March 1, 1940 and the time of enumeration ("school" in IPUMS extract).
- **Children working, 1910-1940 (census)** Individual between the ages of 14 and 18 that works for pay (not missing "occ1950" in IPUMS extract).
- **Years of education, 1940 (census).** Calculated from highest year of school or degree completed ("educd" in IPUMS extract)
- **High-school, 1940 (census).** Individual reports completing 12 or more years of education ("educd").
- **College, 1940 (census).** Individual reports completing 16 or more years of education("educd").

School Enrollment at Age

I construct six binary indicators for enrollment at ages 12, 14, 16, 18, 20, and 22 by combining IPUMS school enrollment records across the 1910–1930 census decades with completed years of education reported in the 1940 census. Ages at each census are computed from birth date on the Dawes Roll.

Because allottees are observed only at decadal census intervals, direct observation at each target age is rare. I therefore apply a monotone-inference rule: school attendance is treated as a single uninterrupted spell, so that:

- an allottee observed in school at age X is coded as enrolled at all target ages $\leq X$;
- an allottee observed not in school at age X is coded as not enrolled at all target ages $\geq X$.

Where census observations fall entirely outside an individual's school-age years, completed years of education from 1940 extend the inference. Assuming school entry at age 6, an allottee reporting N years of completed education is inferred to have last been enrolled at age $6 + N$, and target ages are assigned accordingly.

Migration

- **Urban, 1910-1940 (census).** Whether household was located in an urban or rural enumeration district ("Urban" in IPUMS extract). In these census decades, the Census defined as urban "cities and places with more than 2,500 inhabitants... Also includes townships and other political subdivisions (not incorporated municipalities) with a total population of 10,000 or more and a population density of 1,000 or more per square mile."
- **Moved states.** This variable is equal to 1 if there is evidence that this individual moved outside of Oklahoma by 1942 or earlier. I code this outcome variable according to the following sequence:
  1. If the individual was observed in the 1940 census, then this variable is equal to 1 if they lived outside of Oklahoma, and 0 if they lived inside Oklahoma ("stateicp" is equal to 53 in IPUMS data).
  2. If a registration card for this individual for the World War II draft is observed, then this variable is recoded as 1 if the registration took place outside of Oklahoma. This variable is recoded as 0 if the registration took place in Oklahoma only if the individual was not observed in the 1940 census.
  3. If a post-1940 death record is observed, then this variable is recoded as 1 if the death took place outside of Oklahoma. This variable is recoded as 0 if the death took place in Oklahoma only if the individual was not observed in the 1940 census or World War II draft.
  4. If the individual was not observed in the 1940 census, the WWII draft or a death record, but they were observed in the 1930 census, this variable is recoded as 1 if they lived outside of Oklahoma ("stateicp" is not equal to 53 in IPUMS data). I do not recode this variable as 0 based on 1930 census location because I do not know if the individual moved after 1930.

- **Moved counties.** This variable is equal to 1 if there is evidence that this individual moved out of the county in which their Dawes enrollment took place. To generate this variable, I first placed all allotments into one of the eight counties inside the Muscogee (Creek) Nation: Creek. Tulsa, Wagoner, Muskogee, Okmulgee, McIntosh, Okfuskee and Hughes.[29] I then code this outcome variable in a fashion analogous to the "moved states" variable.
- **Moved populated places, 1940 (census).** Enumeration districts, the geographical units over which US censuses of population were taken from 1880 through 1940 ("enumdist" in IPUMS), were not consistent across decades. To quantify sub-county migration, I constructed a crosswalk of geographically consistent "populated places" (PPs) within the Muscogee (Creek) Nation boundaries.
  1. I began with the roughly 114 populated places, as defined by Ancestry.com, into which all 1940 enumeration districts within the Muscogee (Creek) Nation can be classified.
  2. Working backward, I matched each 1930, 1920, and 1910 enumeration district to its corresponding 1940 PP, guided by the populated place labels Ancestry.com assigns to those earlier censuses. The crosswalk is restricted to populated places in which at least one Creek Freedman was observed.

  This outcome variable equals 1 if an individual's 1940 PP differs from their 1910 PP. Because fewer than six percent of linked individuals resided outside the Muscogee (Creek) Nation in 1910, and I do not construct PPs outside the Nation's boundaries, the variable is also equal to 1 for anyone living outside the Nation in 1940, as such individuals have necessarily left their 1910 PP. With these smaller units, I am able to quantify migrations smaller than cross-county, like moving from a rural area of Tulsa county, to Tulsa city proper, for example.

Fertility & Household Size

- **Number of non-Creek offspring by age X (census).** The cumulative count of non-Creek children–defined as in the intergenerational sample construction above–for whom the enrolled Creek Freedman is identified as the biological mother or father via the IPUMS pointer variables `momloc` or `poploc` in at least one census year, and whose imputed birth year falls no later than the enrolled parent's estimated birth year plus X years. The parent's birth year is the median value reported across all census appearances; age at each child's birth is then child birth year minus parent birth year. Coded missing when the parent's last observed census year predates the year in which they would have reached age X, as the existence of additional children born in unobserved subsequent decades cannot be ruled out.
- **Number children ever born, 1910 and 1940 (census).** Number of children ever born to each woman ("chborn" in IPUMS extract). In 1910, the question was asked of ever woman over age 15 who has

[29]Rogers, Mayes and Seminole counties are also partially in the Muscogee (Creek) Nation.

ever been married. In 1940, this question was asked of sample-line respondents.

- **Number children surviving, 1910 (census).** Number of children surviving to each woman over age 15 who has ever been married ("chsurv" in IPUMS extract)

H.8 Neighborhood-Level Outcome Variables

For the following variables, I only consider individuals living in households; I exclude individuals living in group quarters ("gq" not equal to 1 in IPUMS extract).

Education

- **High-school graduation rate, 1940 (census).** The share of the population, over age 24, who report having completed 12 or more years of education ("educd").
- **College graduation rate, 1940 (census).** The share of the population, over age 24, who report having completed 16 or more years of education ("educd").
- **School attendance, 1940 (census).** Proportion of children between the ages of 5 and 17 who attended school or college any time between March 1, 1940 and the time of enumeration ("school" in IPUMS extract).
- **Teacher education, 1940 (census).** Calculated from highest year of school or degree completed ("educd" in IPUMS extract) for teachers (individuals aged 17– 65, "occ1950" equal to 93 in IPUMS extract) living in enumdist.
- **Teacher relative salary income, 1939 (1940 census).** County-average of earnings ("incwage") for all individuals aged 17–65 with occ1950 value 93, divided by the county average of earnings for all individuals aged 17 -65 with positive earnings.

Labor Markets

- **Employment-to-population ratio, 1940 (census).** Share of men ages 25 to 64 who are employed ("sex" equal to 1, "age" between 25 and 64, and "empstat" equal to 1 in IPUMS extract).
- **Unemployment rate, 1940 (census).**
- **Racial wage gap, 1940 (census).** Mean earnings of Black adult men subtracted from mean earnings of white adult men ("incwage" in IPUMS extract).

Families

- **Rate of single-parent households, 1940 (census).**The share of families ("serial" + "famunit") with children that are single-headed.

- **Poverty rate, 1940 (census)** Following Barrington (1997), each member of a family is assigned an earnings poverty threshold according to gender of household head, number of members, number of children, and whether household is a farm or not.[30] All members of the same family receive the same threshold but different families have different thresholds based on the stated characteristics. This rate represents the share of people whose total family earnings ("fwage1") was below their designated poverty threshold. I exclude families with more than nine members as no poverty threshold was defined for these people.

Segregation and Racial Animus

- **Proportion of neighbors white, 1940 (census).** The percent of people living in enumdist with race equal to white ("race" equal to 100 in IPUMS extract).

[30] I define members of a family using the combination of "serial" and "famunit" in IPUMS, recoding children under 15 "living alone" to he primary family in the household.

Table H.1: Balance on Characteristics in 1910

	(1) Whole Sample	(2) No Oil	(3) Oil	(4) p-Value, Mean Difference [N]
Panel A: Treatment				
Oil on own allotment	0.043	0	1	—
	(0.203)			[6,839]
Panel B: Predetermined outcomes				
Age, in years	25.370	25.373	25.314	0.955
	(17.619)	(17.656)	(16.803)	[6,834]
Male	0.481	0.480	0.505	0.405
	(0.500)	(0.500)	(0.501)	[6,839]
Born in Oklahoma	0.921	0.923	0.899	0.262
	(0.269)	(0.267)	(0.302)	[2,915]
Mother born in Oklahoma	0.753	0.749	0.821	0.034
	(0.431)	(0.434)	(0.384)	[2,915]
Father born in Oklahoma	0.673	0.672	0.696	0.512
	(0.469)	(0.470)	(0.461)	[2,915]
Number of enrolled children	3.961	3.942	4.215	0.449
	(2.807)	(2.783)	(3.120)	[924]
Panel C: Homeownership				
Family owns home	0.765	0.766	0.750	0.629
	(0.424)	(0.423)	(0.434)	[2,915]
Among owners: paid off	0.748	0.750	0.714	0.368
	(0.434)	(0.433)	(0.454)	[2,231]
Farm	0.708	0.708	0.714	0.855
	(0.455)	(0.455)	(0.453)	[2,915]
Panel D: Occupation				
White-collar	0.031	0.031	0.036	0.810
	(0.174)	(0.173)	(0.187)	[1,118]
Blue-collar	0.310	0.315	0.250	0.214
	(0.463)	(0.465)	(0.436)	[1,118]
Farming	0.658	0.654	0.714	0.261
	(0.474)	(0.476)	(0.454)	[1,118]
Panel E: Education				
Literate	0.796	0.792	0.851	0.094
	(0.403)	(0.406)	(0.357)	[2,158]
Child in school	0.854	0.852	0.900	0.304
	(0.353)	(0.355)	(0.303)	[1,331]
Child works	0.384	0.389	0.304	0.421
	(0.487)	(0.488)	(0.470)	[365]
Panel F: Parental characteristics				

Table H.1: Balance on Characteristics in 1910 *(continued)*

	Whole Sample	No Oil	Oil	p-Value, Mean Difference [N]
Father white-collar	0.047	0.047	0.055	0.787
	(0.212)	(0.211)	(0.229)	[1,127]
Father blue-collar	0.122	0.120	0.164	0.340
	(0.328)	(0.326)	(0.373)	[1,127]
Father farming	0.831	0.833	0.782	0.324
	(0.375)	(0.373)	(0.417)	[1,127]
Father literate	0.819	0.823	0.745	0.147
	(0.385)	(0.382)	(0.440)	[1,127]
Mother literate	0.786	0.787	0.782	0.932
	(0.410)	(0.410)	(0.417)	[1,077]
Panel G: County of allotment				
Wagoner County	0.173	0.178	0.051	0.000
	(0.378)	(0.383)	(0.221)	[6,839]
Okmulgee County	0.169	0.160	0.365	0.000
	(0.375)	(0.367)	(0.482)	[6,839]
Creek County	0.166	0.165	0.205	0.070
	(0.372)	(0.371)	(0.404)	[6,839]
Muskogee County	0.164	0.161	0.232	0.001
	(0.370)	(0.367)	(0.423)	[6,839]
Okfuskee County	0.076	0.078	0.020	0.000
	(0.264)	(0.268)	(0.142)	[6,839]
Other county	0.252	0.258	0.126	0.000
	(0.434)	(0.437)	(0.333)	[6,839]

Note: This table presents balance tests on pre-treatment characteristics of Creek Freedmen allottees, by whether oil was discovered on the allottee's own allotment (Panel A). For each variable, the top row reports p-value from a two-sample t-test of the difference in means between the No Oil and Oil groups; the row below reports the standard deviation (in parentheses) and the number of non-missing observations (in brackets). Panel A reports oil's sample mean, with the No Oil and Oil columns fixed at 0 and 1 by construction. Age and gender (Panel B) are observed for the full allottee sample; all remaining variables are restricted to allottees linked to the 1910 Census, and further restricted by cohort, where a variable is defined only for Older Allottees (occupation, Panel D) or Younger Allottees (school attendance, child labor, and parental characteristics, Panels E–F) in 1910. Panel G reports the share of allottees whose own allotment was located in each of the five largest counties in the sample, with all remaining counties grouped as "Other." For variable definitions and sample construction, see Data Appendix H.

I Additional Balance Tests and Robustness

This appendix provides tests for statistical balance over observables as well as further robustness checks on the main findings. I begin with a series of supplementary tests that demonstrate further that the discovery of oil was not associated with many characteristics that we might consider important. I then consider several threats to identification and either replicate my main results over carefully chosen alternative samples or otherwise provide additional evidence.

I.1 Balance

In this subsection, I present two sets of balance tests: whether the incidence of oil discovery differed by allottee race, and whether allottees whose land later produced oil differed from those whose land did not on individual- and household-level characteristics measured prior to drilling.

In **Table I.18**, I establish that oil-producing wells are built on the allotments of Creek Freedmen and Creek Indians at statistically identical rates.

I.2 Positive Selection in Drilling

The results above could in principle be confounded if allottees who happened to receive higher-quality land — land later selected for drilling — were also more skilled or ambitious, biasing comparisons between treated and untreated allottees more broadly. To address this, I restrict the comparison group to allottees with a *dry well* drilled on their land: speculators judged these parcels promising enough to drill, but the well did not produce. Comparing producing-well allottees to dry-well allottees isolates the effect of oil wealth from selection on land quality or personal acumen. The following tables reproduce the main results within this restricted sample.

Across this restricted sample, the qualitative pattern of the main results holds: producing-well allottees show higher school enrollment during their teenage years and are less likely to remain in their original township relative to dry-well allottees, and the education gains reported in the neighborhood-quality table persist. Point estimates on migration, homeownership, and fertility are directionally similar to the main results but imprecisely estimated, reflecting the much smaller sample of allottees with a dry well on their allotment.

I.3 Positive Selection in Linkage Methods

Linking allottees to census records combined an automated matching algorithm with hand review of ambiguous cases. If the allottees I was able to match by hand differ systematically from those matched automatically, this could bias the main estimates. The following tables re-estimate the main specifications restricting the sample to automated links only, excluding all hand-linked matches. The core occupational and urbanization results survive this restriction. The white-collar effect remains positive, statistically significant, and of similar magnitude (**Table** I.28); treated allottees remain significantly more likely to live in urban areas, driven as in the main sample by those who moved (**Table** I.32); the homeownership estimates remain null (**Table** I.33 and **Table** I.34); and the neighborhood school-attendance advantage is essentially unchanged (**Table** I.35). A second group of results keeps the same sign and similar magnitude but loses statistical significance in the smaller sample: the migration coefficients (**Table** I.31), the marriage catch-up for men at age 24 (**Table** I.29), the pooled effect of parental treatment on children's school attendance (**Table** I.30), and the whiter-neighborhoods coefficient (**Table** I.35). The remaining results change in more than precision. In **Table** I.30, the offspring counts, null in the full sample, turn negative and significant here—the pooled estimates at ages 30 and 40 and the fathers' estimate at age 40. This is to be expected, because, by construction, this table includes more treated folks who are "harder to find." The mother-specific effect on children's school attendance loses significance. In **Table** I.32, the negative stayer coefficients of the main sample fall to approximately zero. And two sets of cells in **Table** I.29 cannot be informatively estimated in the automated-only sample and are suppressed: the school-enrollment cells at the school-leaving ages, where the treated group retains no outcome variation—every treated man observed at age 16, and every treated woman observed at age 16 or 18, remains enrolled—and the marriage and household-formation outcomes for women.

MISS BAYONNE AVONNE DAVIS

Miss Davis is the beautiful and accomplished daughter of Mr. and Mrs. George Davis, Boynton, Oklahoma. The little Miss is a graduate of Wilberforce, Class of '28, and is also a member of the Alpha Kappa Alpha Chapter.

Whether to teach school this year or go to work as cashier for the Security Life Insurance Co., is what is puzzling the head of Miss Davis these days. Her father is general manager of Oklahoma's wide awake insurance company, the Security Life.

"I may work for Dad," said Miss Davis when urged to announce her decision about the two jobs.

Figure I.1: *The Black Dispatch*, August 16, 1928

WESTERN UNIVERSITY.

Last Friday evening a mock trial was the program of the James A. Handy literary society.

Miss Eva Jones of Denver gave the fourth of the series of six recitals by the advance piano students under Prof. Robert Jackson. Miss Jones has made wonderful advancement, as interpreted by the many difficult high class numbers rendered with such ease and beauty on this occasion.

Miss Anna Rentie, a student of the Business department, returned Friday from a two weeks sight seeing visit to Denver, Colorado Springs and Salt Lake City, Utah, with her father, Mr. Ireland Rentie of Muskogee, Okla., and her sister, Miss Mattie Rentie. Mr. Rentie is a very wealty negro land leasor, receiving a large income monthly from the Standard Oil company of New York. On this entire journey Mr. Rentie and his daughters occupied a private Pullman car specially chartered for this trip. He will return for our commencement.

Mr. Eugene Vaughan of last year's class in stenography, who was recently sent to fill a lucrative position in the office of superintendent of insurance at

Figure I.2: *The Topeka Plaindealer*, May 8, 1908

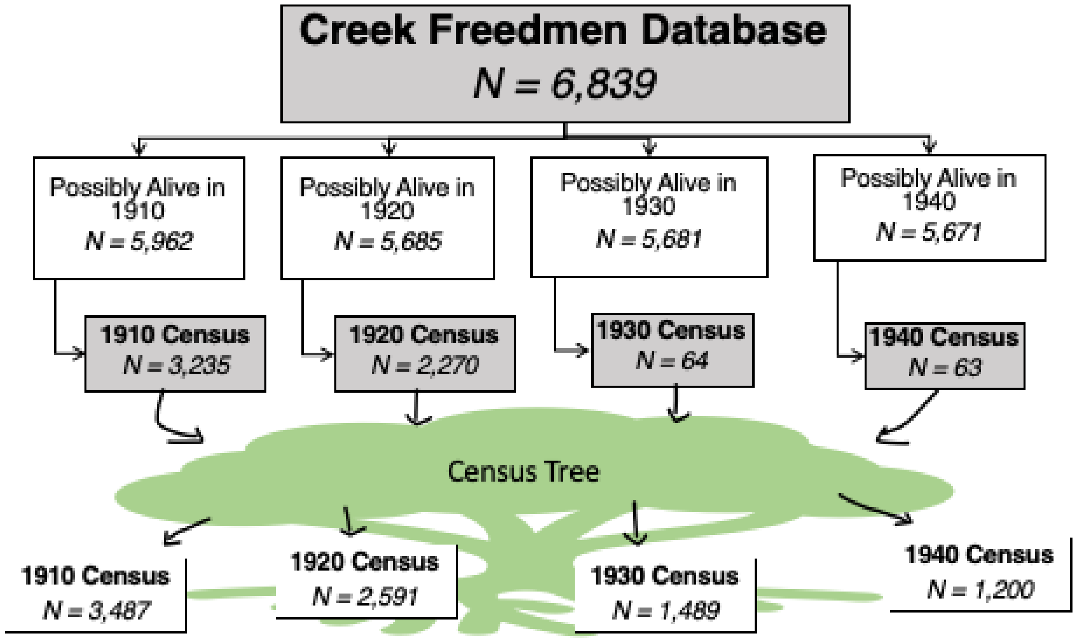


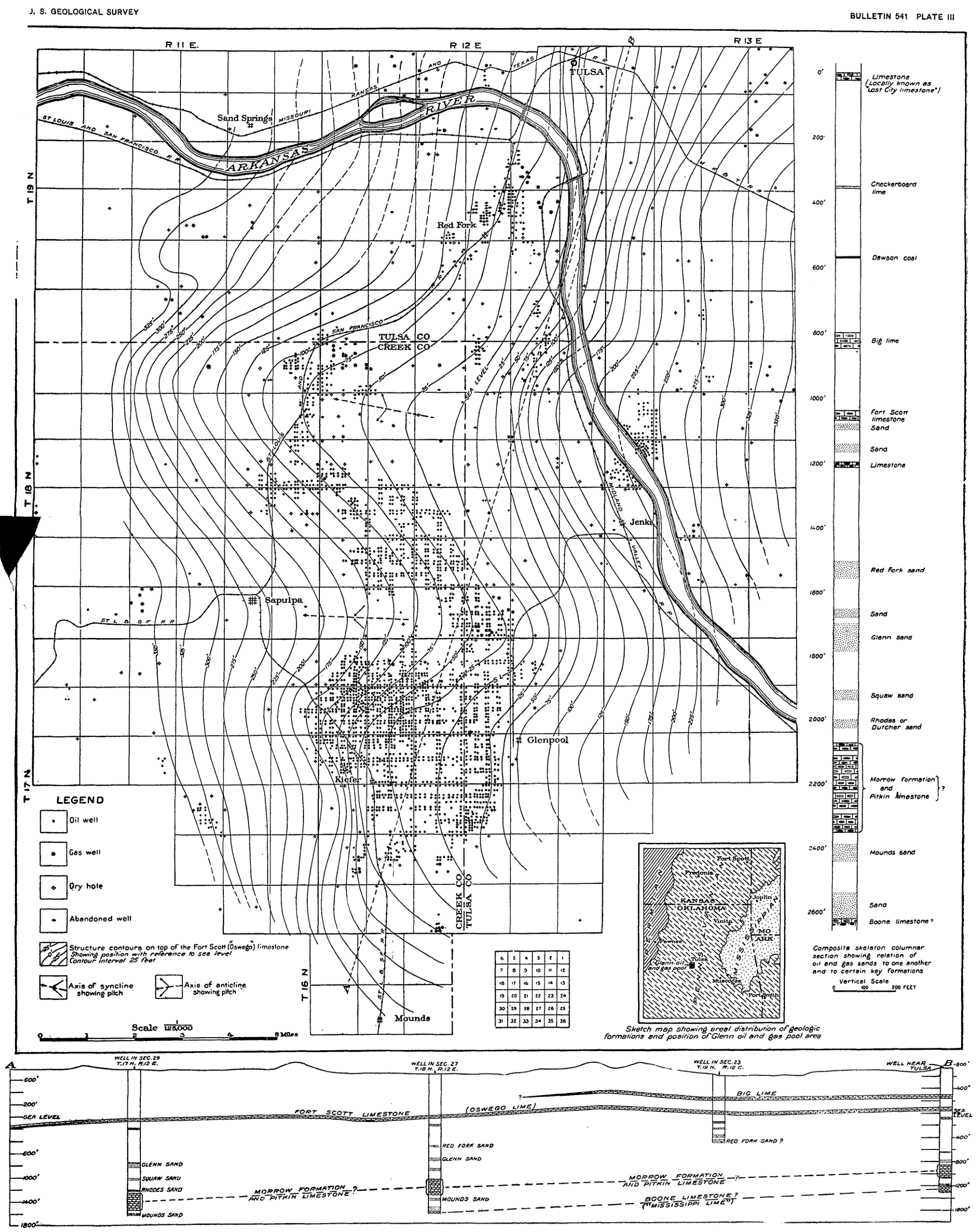
Figure I.3: Creek Freedmen Allottee Data Linking

Figure I.4: The New Oil Lease Regulations, *The Tulsa Weekly Democrat*, June 28, 1907

THE NEW OIL LEASE REGULATION CHANGES

THE FOLLOWING ARE THE NEW LEASE REGULATIONS MADE EFFECTIVE BY THE DEPARTMENT OF TH INTERIOR. THERE ARE MANY IMPORTANT CHANGES. THIS SET EMBRACES ALL PREVIOUS REGULATIONS THAT WERE NOT CHANGED, AND A LESSEE MUST FOLLOW THE NEW ONES TO THE LETTER ON ALL LEASES THAT ARE TO BE APPROVED BY THE SECRETARY OF INTERIOR. LEASE MEN, PRODUCERS, ALLOTTEES AND ATTORNEYS WILL FIND MANY IMPORTANT CHANGES IN THE NEW REGULATIONS THAT ARE OF VITAL IMPORTANCE TO THEM. THIS IS THE FIRST AND ONLY PUBLICATION IN THE TERRITORY AND THE REGULATIONS ARE OFFICIALLY GIVEN OUT BY INDIAN AGENT KELSEY.

Figure I.5: The Glenn Oil and Gas Pool and Vicinity, Oklahoma, Contributions to Economic Geology, 1912, Part II



MAP OF GLENN OIL AND GAS POOL AND VICINITY, OKLAHOMA.

By Carl D. Smith.

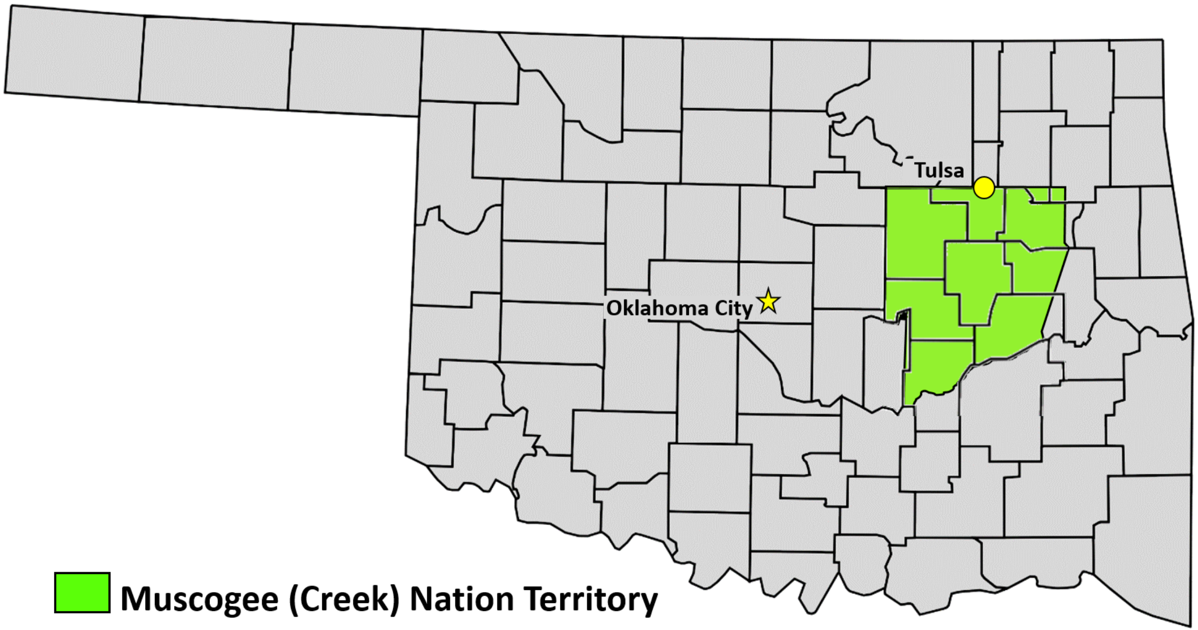


Figure I.6: Muscogee (Creek) Nation Territory within an Oklahoma county map

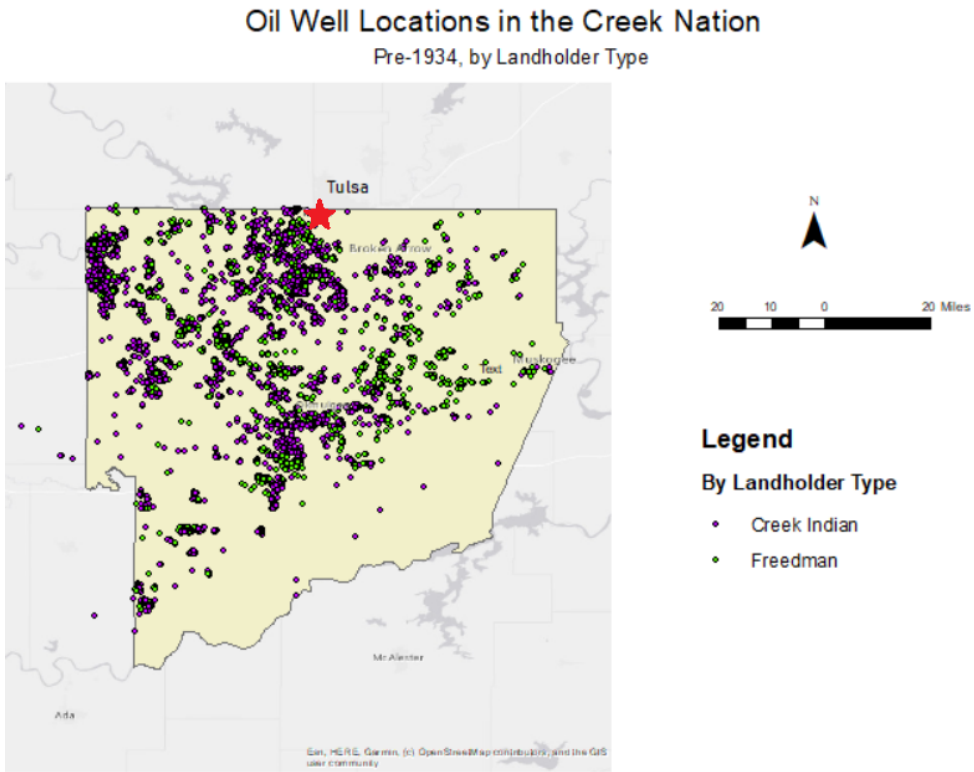


Figure I.7: Oil Well Locations in the Muscogee (Creek) Nation

Note. This figure displays a map of the Muscogee (Creek) Nation with producing oil wells plotted by the race of the landholder.

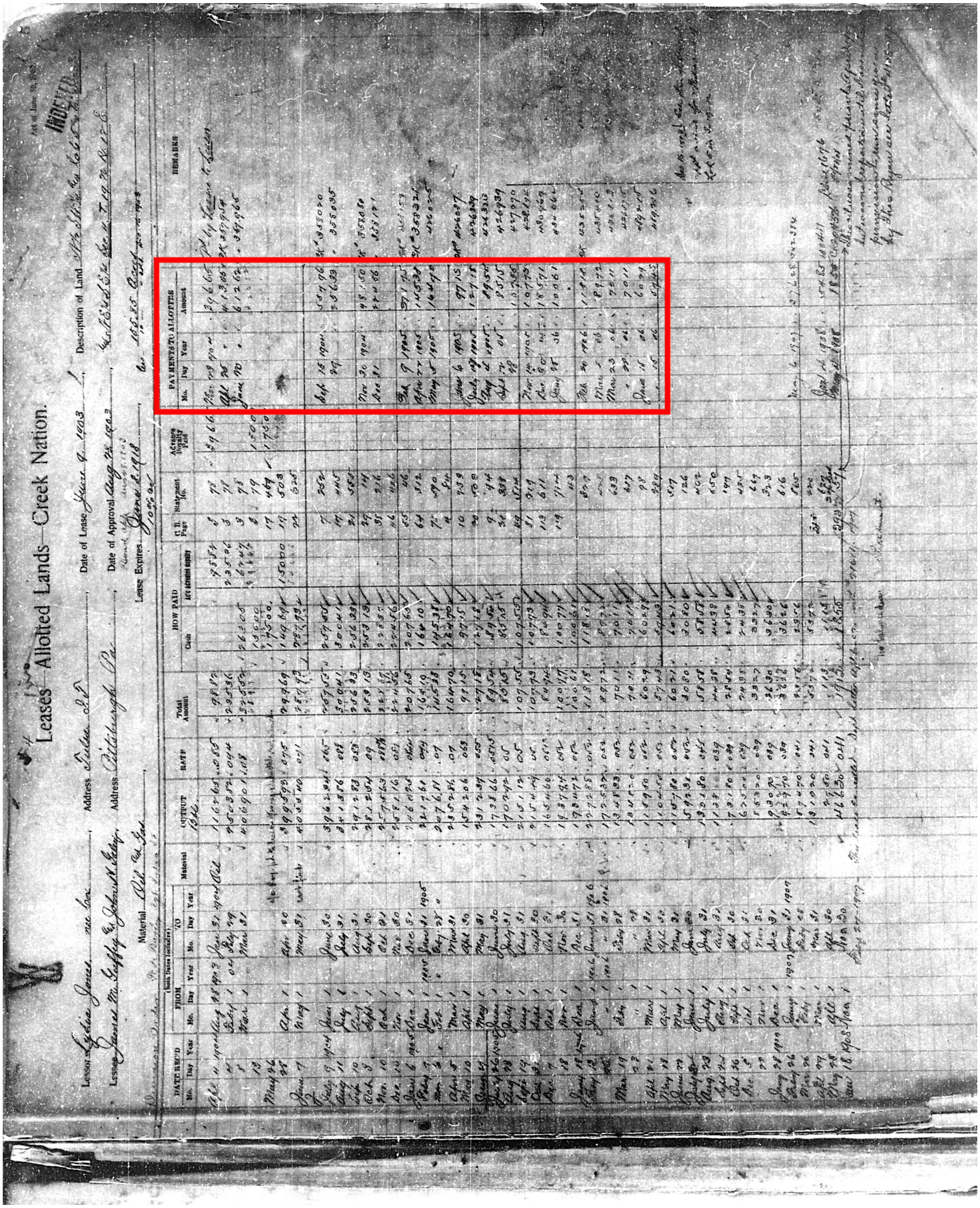
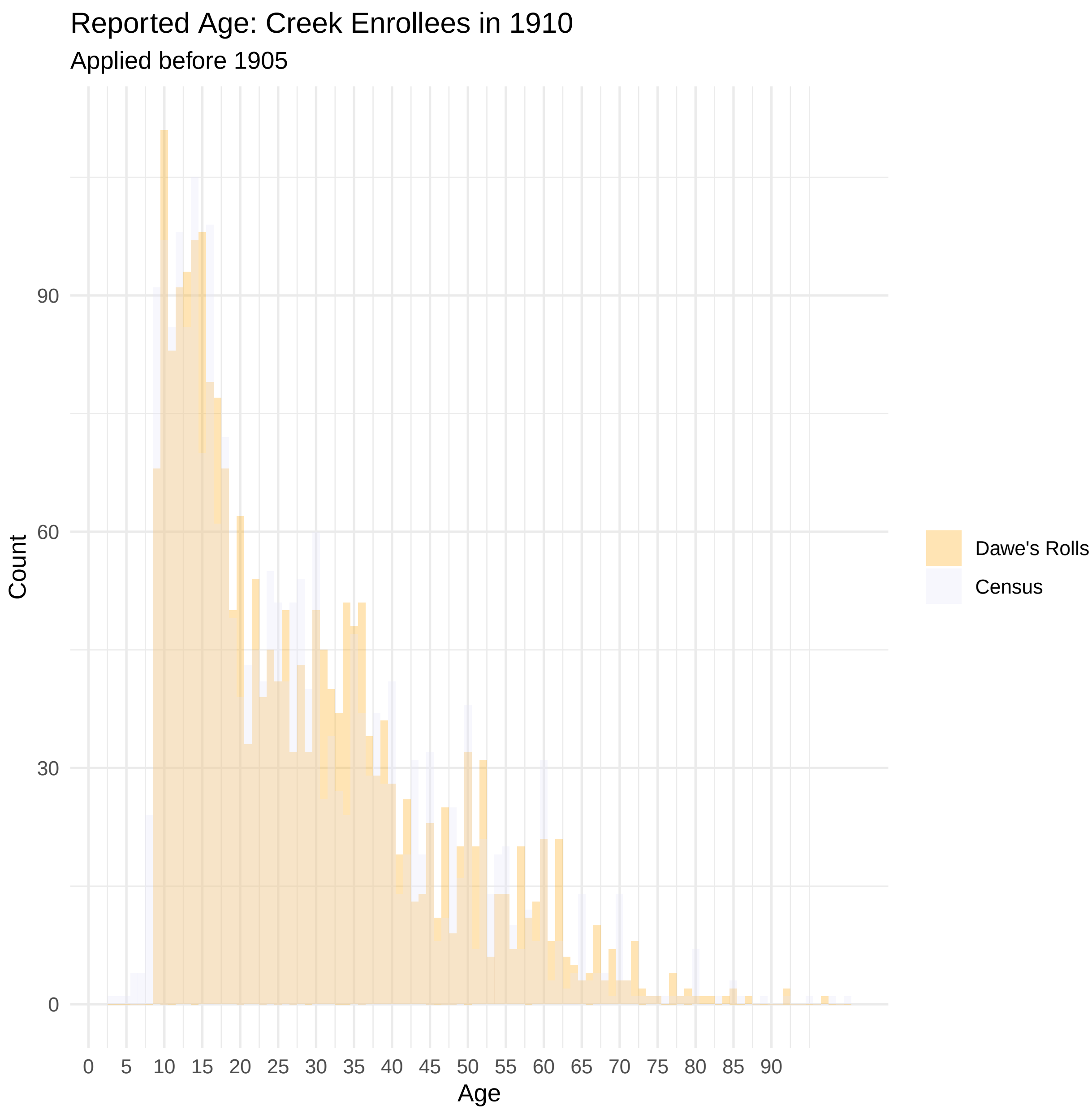


Figure I.8: Oil Lease, 1904

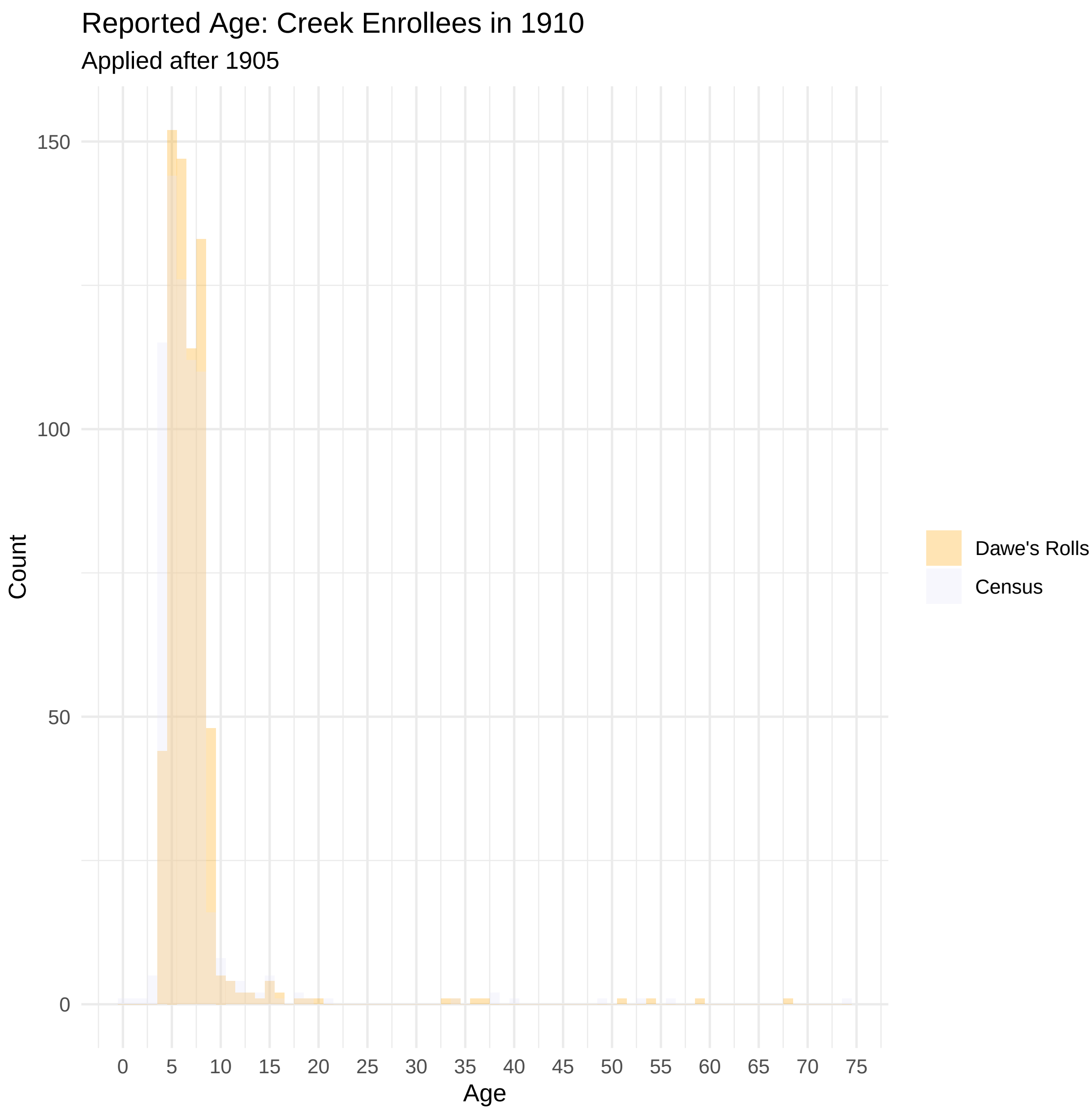
Note. We can see from this lease that, between March 1904 and May 1908, Creek Native Lydia Jones nee Cox received a total of $5,440.65 in royalty payments from James M. Guffey and John N. Galey of Pittsburgh, PA.

Figure I.9: Bunching in Creek Freedmen database versus 1910 Census.



Note: Laws regarding dates of birth and death to qualify for enrollment as a Creek Freedman are detailed in Subsection E.2.

Figure I.10: Bunching in Creek Freedmen database (Minors and Newborns) versus 1910 Census



Note: Laws regarding dates of birth and death to qualify for enrollment as a Creek Freedman Minor or Newborn are detailed in Subsection E.2.

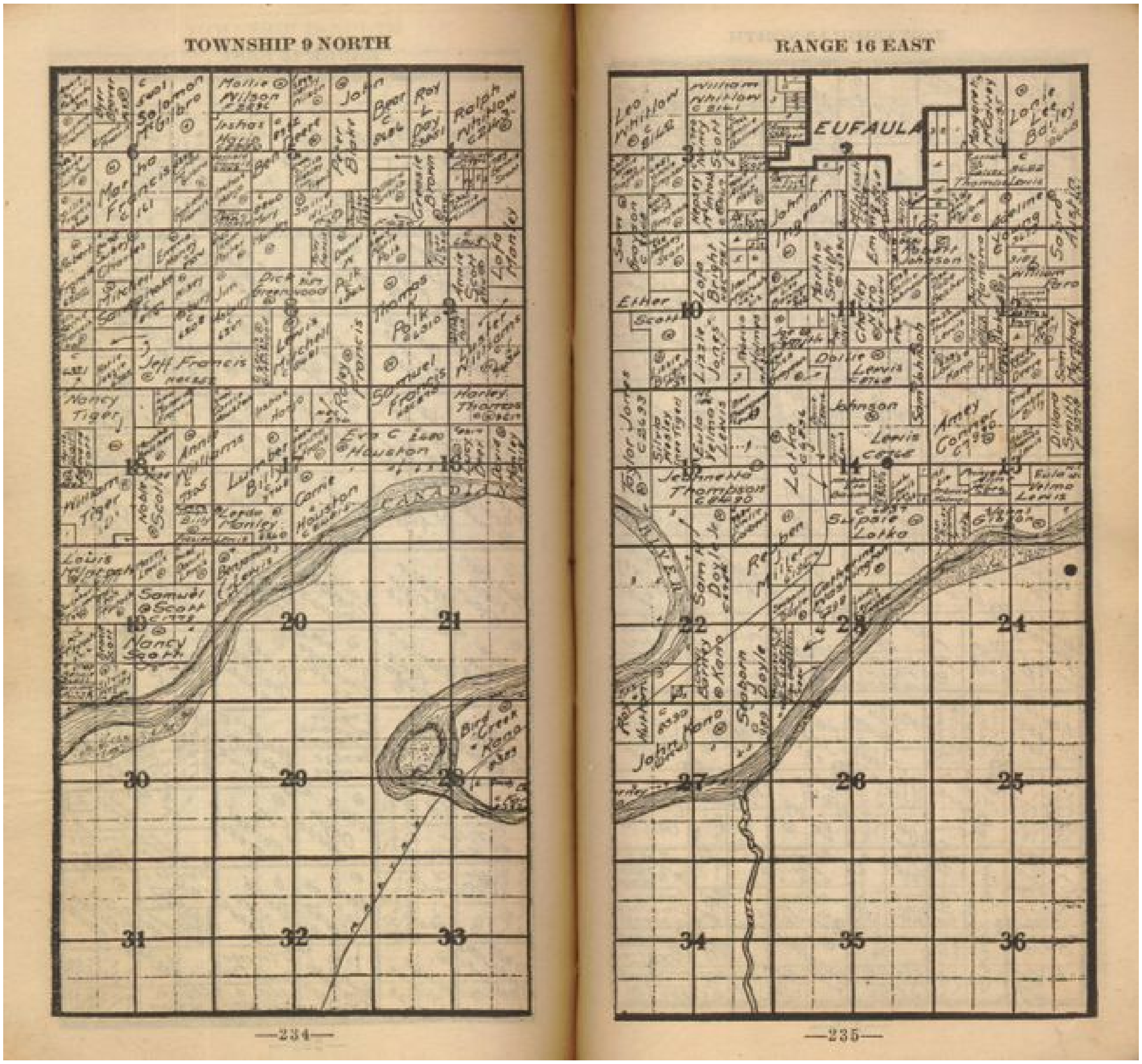


Figure I.11: Township 9 North, Range 16 East

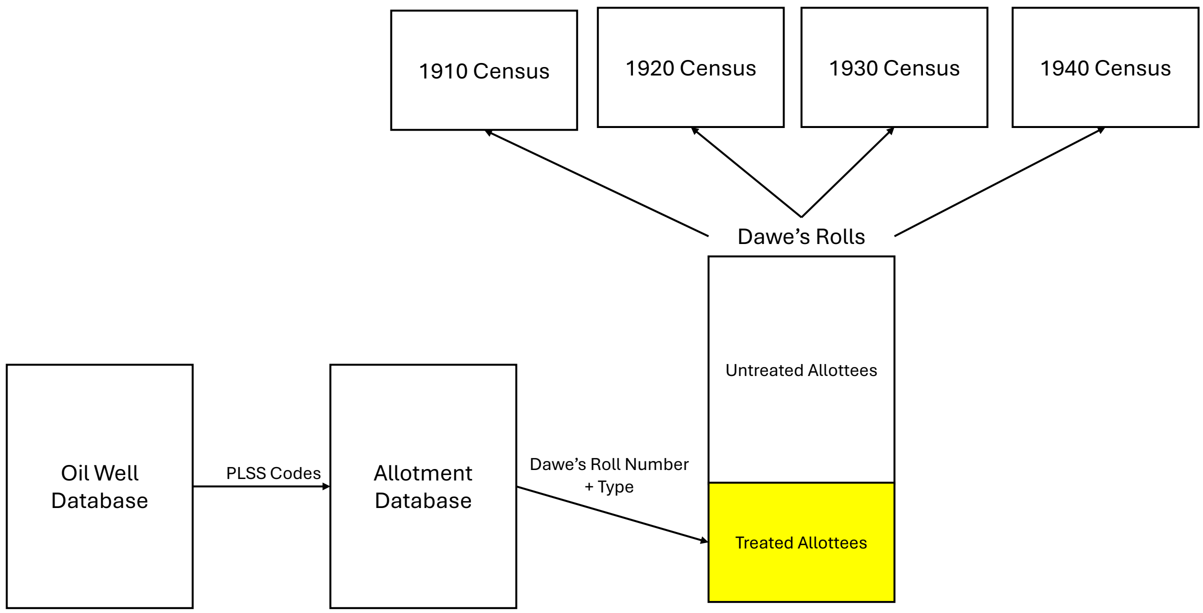


Figure I.12: Treatment Assignment

Table I.1: The Effect of Oil Money on Self-Employment

	Older Allottees	Younger Allottees
	(1)	(2)
Oil	-0.058	0.041
	(0.056)	(0.052)
N	1566	1550
Mean	0.654	0.290

Note: Each column presents a separate OLS regression. The sample consists of Creek Freedmen matched to the 1920, 1930, or 1940 federal census. The outcome equals one if self or husband are self-employed and zero if self or husband works for wages; individuals not in the labor force are excluded. Each observation is an individual-census-year pair. Older refers to individuals born in 1892 or earlier; Younger refers to individuals born after 1892 (under age 18 in 1910). Oil indicates that at least one of the individual's allotted plots contained an oil well by the end of 1918. All specifications include year fixed effects and control for age in 1910, age in 1910 squared, and an indicator for male. The mean of the dependent variable is reported for each subsample. Standard errors are clustered at the family level. *$p < 0.10$, **$p < 0.05$, ***$p < 0.01$.

Table I.2: Transitions into and out of Homeownership

	Homeowner to Non-homeowner Mean: 0.44	Non-homeowner to homeowner Mean: 0.42
year=1920 X Oil by 1918	0.042	0.127
	(0.049)	(0.092)
year=1930 X Oil by 1918	-0.127*	-0.110
	(0.072)	(0.107)
year=1940 X Oil by 1918	-0.087	0.055
	(0.074)	(0.145)
Controls (age, age^2, gender)	Yes	Yes
Decade fixed-effects	Yes	Yes
Adjusted R-squared	0.10	0.02
Observations	2,844	835

Note: This table presents a series of regression estimates of the difference in likelihood of transitioning into or out of home ownership, respectively, between treated and untreated Creek Freedmen allottees in 1920, 1930 and 1940. I define original homeownership status as homeownership status in 1910, if I observe them in the 1910 census. All regressions control for age, age squared, and gender and include fixed effects for census decade. Means are computed over the entire treated and untreated Creek Freedmen allottee population in 1920. Treatment is defined as presence of a producing oil well on own allotment by the end of 1918. For more information on data and variable definitions, see Data Appendix H. Standard errors are clustered at the Family ID level and are shown in parentheses. ***$p < 0.01$,**$p < 0.05$,*$p < 0.1$

Table I.3: The Effect of Oil Money on Migration: Ordered Logit Results

	Pooled 1920–1940
Oil	-0.253
	(0.174)
Thresholds	
Stayed in county | Left county	-1.140
	(0.249)
Left county | Left Creek Nation	0.890
	(0.232)
Left Creek Nation | Left Oklahoma	1.623
	(0.237)
Observations	4,261

Note: The outcome is an ordered categorical variable coded 0 (remained in allotment county), 1 (left allotment county but remained within the Creek Nation), 2 (left the Creek Nation but remained in Oklahoma), and 3 (left Oklahoma). Estimated by ordered logit (proportional odds). The sample pools observations from the 1920, 1930, and 1940 censuses for Creek allottees observed in one or more census. Oil is an indicator for whether the allotment parcel contained oil. All specifications control for age in 1910, age in 1910 squared, a male indicator, and year fixed effects. Standard errors (in parentheses) are clustered by individual. Thresholds are the estimated cut-points on the latent mobility scale separating adjacent outcome categories. $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

Table I.4: The Effect of Oil Money on School Attendance for Younger Allottees, by Variable Source

	Oil	Mean	N
Panel A: Census observations only			
Age 12	-	0.99	976
Age 14	-	0.98	815
Age 16	0.093**	0.86	557
	(0.035)		
Age 18	0.247**	0.49	382
	(0.110)		
Age 20	0.093	0.12	342
	(0.132)		
Age 22	-	0.06	417
Panel B: 1940 reported education only			
6 years	-	0.99	402
8 years	-	0.95	313
10 years	0.225*	0.65	221
	(0.113)		
12 years	0.209	0.29	269
	(0.153)		
14 years	0.258*	0.07	324
	(0.142)		
16 years	0.184*	0.03	407
	(0.098)		

Note: Sample is Creek Freedmen allottees born after 1892, matched to at least one census decade. Dependent variables are binary indicators for school enrollment at the indicated threshold. Panel A constructs indicators via monotone inference from IPUMS enrollment records across the 1910–1930 censuses: an allottee observed in school at age X is coded as enrolled at all target ages $\leq X$; one observed not in school at age X is coded as not enrolled at all target ages $\geq X$. Row labels indicate the enrollment age threshold. Panel B uses years of education reported in the 1940 census to construct the dependent variable, assuming school entry at age 6 (maximum schooling age $= 6 +$ years of education); row labels indicate the years-of-education threshold. All specifications control for age in 1910 (linear and quadratic) and sex. Standard errors clustered by family in parentheses. Treatment is defined as the presence of a producing oil well on own allotment by the end of 1918. For more information on data and variable definitions, see Data Appendix H. Dashes mark cells where the outcome does not vary among treated allottees at the indicated age; the linear probability model is uninformative at that boundary, so the coefficient and standard error are suppressed while the mean and sample size are retained. ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$

Table I.5: Neighborhood Quality for Allottees , 1940

	Estimate	Mean	Observations
A. Education			
Share graduated college	0.025	0.14	583
	(0.032)		
Share graduated high school	0.031	0.29	583
	(0.030)		
School attendance	0.018**	0.83	583
	(0.008)		
Teacher Salary	50.698	980.02	535
	(47.449)		
B. Labor Markets			
Employment to population ratio	0.027**	0.80	583
	(0.014)		
Unemployment rate	-0.024**	0.13	582
	(0.011)		
Racial earnings gap	6.751	537.46	473
	(55.823)		
C. Families			
Share of families single-parent	0.006	0.18	581
	(0.012)		
Poverty rate	-0.043*	0.67	583
	(0.025)		
D. Segregation			
Proportion of neighbors white	-0.059	0.57	583
	(0.041)		
E. Other			
Urban	-0.001	0.52	587
	(0.057)		

Note: This table presents regression estimates of selected neighborhood characteristics of the places where treated and untreated Creek Freedmen allottees born after 1892 lived in 1940. All regressions control for age, age squared, and gender. Means are computed over the full treated and untreated Creek Freedmen allottee population. Treatment is defined as the presence of a producing oil well on own allotment by the end of 1918. For more information on data and variable definitions, see Data Appendix H. Standard errors are clustered at the Family ID level and are shown in parentheses. ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$

Table I.6: White-Collar Jobs in 1910

	Percent
Occupation, 1950 basis	
Clergymen	3.85
Nurses, professional	3.85
Teachers (n.e.c.)	15.38
Managers, officials, and proprietors (n.e.c.)	53.85
Collectors, bill and account	3.85
Mail carriers	3.85
Messengers and office boys	3.85
Real estate agents and brokers	3.85
Salesmen and sales clerks (n.e.c.)	7.69
Total	100.00

Note: Distribution of white-collar occupations among Creek Freedmen allottees in 1910, classified using 1950 census occupational codes.

Table I.7: White-Collar Jobs in 1920

	Percent
Occupation, 1950 basis	
Clergymen	8.11
Lawyers and judges	8.11
Nurses, professional	2.70
Social and welfare workers, except group	2.70
Teachers (n.e.c.)	10.81
Managers, officials, and proprietors (n.e.c.)	18.92
Bookkeepers	5.41
Cashiers	2.70
Mail carriers	2.70
Shipping and receiving clerks	2.70
Telephone operators	2.70
Clerical and kindred workers (n.e.c.)	2.70
Demonstrators	2.70
Newsboys	2.70
Real estate agents and brokers	13.51
Salesmen and sales clerks (n.e.c.)	10.81
Total	100.00

Note: Distribution of white-collar occupations among Creek Freedmen allottees in 1920, classified using 1950 census occupational codes.

Table I.8: White-Collar Jobs in 1930

	Percent
Occupation, 1950 basis	
Athletes	1.72
Clergymen	3.45
Lawyers and judges	1.72
Librarians	1.72
Pharmacists	1.72
Teachers (n.e.c.)	17.24
Postmasters	1.72
Managers, officials, and proprietors (n.e.c.)	32.76
Messengers and office boys	1.72
Shipping and receiving clerks	1.72
Stenographers, typists, and secretaries	5.17
Clerical and kindred workers (n.e.c.)	5.17
Advertising agents and salesmen	1.72
Insurance agents and brokers	1.72
Real estate agents and brokers	12.07
Salesmen and sales clerks (n.e.c.)	8.62
Total	100.00

Note: Distribution of white-collar occupations among Creek Freedmen allottees in 1930, classified using 1950 census occupational codes.

Table I.9: White-Collar Jobs in 1940

	Percent
Occupation, 1950 basis	
Clergymen	4.26
Librarians	2.13
Musicians and music teachers	6.38
Nurses, professional	2.13
Pharmacists	4.26
Social and welfare workers, except group	2.13
Surveyors	2.13
Teachers (n.e.c.)	12.77
Therapists and healers (n.e.c.)	2.13
Inspectors, public administration	2.13
Managers, officials, and proprietors (n.e.c.)	21.28
Clerical and kindred workers (n.e.c.)	12.77
Insurance agents and brokers	2.13
Real estate agents and brokers	8.51
Salesmen and sales clerks (n.e.c.)	14.89
Total	100.00

Note: Distribution of white-collar occupations among Creek Freedmen allottees in 1940, classified using 1950 census occupational codes.

Table I.10: Creek Indian and Freedmen Royalty Accounts

	Total Credits Received 1908-1918				
	# Accounts	Mean	Median	Minimum	Maximum
Number of Quarters Paid					
0	2	0	0	0	0
1	1,974	47.90655	24	.38	1981
2	2,487	121.1301	52	1.5	5864.83
3	1,878	229.3537	104	3	56546.07
4	1,348	435.1282	174	3.74	39840.6
5	953	639.0257	266.24	1.9	24063.24
6	810	1127.39	348	2.28	72124.1
7	546	1626.822	556.095	6.75	347497.7
8	432	3908.359	719.88	13	470667.3
9	320	8264.683	870.975	9	771851.1
10	282	14050.53	1048	15	1604416
11	179	18128.07	988	22	868537.2
12	162	12163.8	1068	22	360807.3
13	148	7946.048	1062.065	13	449181
14	101	4148.589	1728	14.14	57775.68
15	87	6824.173	1450.96	126.88	86824.3
16	62	4009.014	1183.08	19.2	47456.29
17	62	3547.767	2115	22	27986.78
18	38	3726.95	1768.135	27	32718.36
19	65	8770.296	580	42	235177.2
20	22	4686.077	1426.365	36.96999	36221.45
21	40	6201.217	1636.95	36	102185.3
22	21	12422.9	3440.94	27	65336.92
23	23	9614.618	2382	33	65613.92
24	26	9239.475	1255.36	144	68585.37
25	11	6616.052	6467.96	145.5	15258.04
26	18	6693.16	3784.52	392	53158.2
27	16	18848.94	3224.995	65.75	92671.73
28	10	37519.75	15453.56	336	241168
29	12	11134.49	2160.135	174	53151.16
30	54	1963.252	569.3	51	12536.6
Total	12,189	1877.621	144	0	1604416

Note: Distribution of total royalty credits received 1908–1918 by number of quarterly payments, for both Creek Indian and Freedmen allottees with royalty accounts.

Table I.11: Freedmen with Royalty Accounts

	Total Credits Received 1908-1918				
	# ALLOTTEES	Mean	Median	Minimum	Maximum
Number of Quarters Paid					
0	6,187	0	0	0	0
1	154	36.26903	12	1.5	1076.02
2	111	125.0732	52	6.65	3393.64
3	77	157.6096	104	8	608
4	58	549.246	138.5	11.07	20484.02
5	39	318.9474	200	10	1038.36
6	30	289.3713	200.55	33.44	1097.24
7	27	683.8044	274.14	42	6984.1
8	26	1033.137	494.185	36.4	12581.92
9	19	1457.378	496	51.08	11254.65
10	20	2112.78	819.5	120	11518.13
11	10	1368.145	480.52	144	6119.68
12	8	2948.536	1393.2	621	10549.58
13	12	984.2358	478	78	3488.35
14	3	893.8667	336	258.05	2087.55
15	6	15426.57	11107.6	183	37762.63
16	3	1275.953	1822	19.2	1986.66
17	2	2028.53	2028.53	1161.65	2895.41
19	11	491.8355	114	46.64001	3109
20	3	991.9601	1222	349.4001	1404.48
21	2	3111	3111	3076	3146
23	1	1908.28	1908.28	1908.28	1908.28
24	5	1770.318	288	177	7810.59
25	1	1582.39	1582.39	1582.39	1582.39
26	3	629.18	739.5399	392	756
27	1	1399.14	1399.14	1399.14	1399.14
28	2	2064.26	2064.26	1358	2770.52
30	18	474.5078	307	51	1789.4
Total	6,839	444.7794	46	0	37762.63

Note: Distribution of Creek Freedmen allottees with royalty accounts by number of quarterly payments received 1908–1918.

Table I.12: Characteristics of Rural and Urban Enumeration Districts

	Rural	Urban	p-value
A. 1920 Census			
Nonwhite home ownership rate	0.40	0.25	0.00
	(0.24)	(0.23)	
Nonwhite white-collar occupation rate	0.04	0.08	0.00
	(0.08)	(0.06)	
Nonwhite child school attendance	0.61	0.75	0.00
	(0.23)	(0.16)	
Literacy among nonwhite children/teenagers	0.91	0.98	0.00
	(0.12)	(0.03)	
B. 1930 Census			
Nonwhite home ownership rate	0.31	0.29	0.19
	(0.20)	(0.23)	
Nonwhite white-collar occupation rate	0.04	0.09	0.00
	(0.07)	(0.08)	
Nonwhite child school attendance	0.72	0.79	0.00
	(0.20)	(0.16)	
Literacy among nonwhite children/teenagers	0.92	0.98	0.00
	(0.14)	(0.05)	
C. 1940 Census			
Nonwhite home ownership rate	0.36	0.23	0.00
	(0.23)	(0.20)	
Nonwhite white-collar occupation rate	0.05	0.11	0.00
	(0.07)	(0.10)	
Nonwhite child school attendance	0.77	0.84	0.00
	(0.22)	(0.16)	
Literacy among nonwhite children/teenagers	0.87	0.92	0.03
	(0.25)	(0.22)	

Note: The table presents descriptive statistics over the enumeration districts in which at least one allottee was found living in each Census year, split by whether the ED was classified as "urban" or "rural" (see Section H). Mean (Standard deviation): p-value from a pooled t-test. To be clear, these statistics consider *all* nonwhite households, not only Creek Freedmen households.

Table I.13: Method with Which Census-Link Made

	Census Decade			
	1910	1920	1930	1940
Linked				
Matched by Hand	300	234	192	152
Semiautomated: Family	2,029	963	—	—
Semiautomated: Unique Name	326	361	—	—
Census Tree (implied)	243	432	1,033	844
Total linked	2,898	1,990	1,225	996
Not linked				
Confirmed deceased	891	1,129	1,141	1,151
Unmatched	3,050	3,720	4,473	4,692
Total	6,839	6,839	6,839	6,839

Note: Creek Freedmen allottees were linked to the various censuses in rounds using various methods. This table breaks down the method with which each link was made, as well as the deceased and unmatched individuals. In cases where the same link was made through multiple methods, it is grouped with the most reliable linking method. Observations of deaths are incomplete, particularly after 1920.

Table I.14: Census-Linkage Rates, by Treatment Status

	No oil by 1918 [6,546]	Oil by 1918 [293]	p-value
Panel A: All links			
Link result, 1910	0.479 (0.500)	0.669 (0.471)	<0.001
Link result, 1920	0.343 (0.475)	0.481 (0.501)	<0.001
Link result, 1930	0.208 (0.406)	0.368 (0.483)	<0.001
Link result, 1940	0.170 (0.375)	0.298 (0.458)	<0.001
Panel B: Excluding hand-linked (Match Method = 1)			
Link result, 1910	0.443 (0.497)	0.307 (0.462)	<0.001
Link result, 1920	0.312 (0.463)	0.205 (0.405)	<0.001
Link result, 1930	0.181 (0.385)	0.180 (0.385)	0.955
Link result, 1940	0.149 (0.356)	0.134 (0.342)	0.522

Note: Mean (Standard deviation), p-value from a pooled t-test. Allottees in this table are grouped by whether oil was produced under their own allotment by the end of 1918 or not. The Ns at the top of the columns represent the entire Creek Freedmen population, but the denominators with which the linkage rates are calculated exclude individuals found to be deceased by the respective census year.

Table I.15: Census-linked sample characteristics

	Unlinked	Linked	p-value
1910 Census			
N	2,458 (41.2%)	3,505 (58.8%)	
Male	0.432 (0.495)	0.519 (0.500)	<0.001
Age in 1910	26.102 (17.346)	23.223 (16.216)	<0.001
1920 Census			
N	3,095 (54.4%)	2,591 (45.6%)	
Male	0.425 (0.494)	0.557 (0.497)	<0.001
Age in 1910	25.429 (17.230)	22.084 (15.012)	<0.001
1930 Census			
N	4,180 (73.6%)	1,502 (26.4%)	
Male	0.420 (0.494)	0.667 (0.471)	<0.001
Age in 1910	24.800 (17.186)	21.398 (13.427)	<0.001
1940 Census			
N	4,462 (78.7%)	1,210 (21.3%)	
Male	0.424 (0.494)	0.708 (0.455)	<0.001
Age in 1910	25.016 (17.169)	19.655 (11.797)	<0.001

Note: Mean (Standard deviation), p-value from a pooled t-test. Allottees in this table grouped by whether linked to each census or not. The linked and unlinked units in this table include both treated and untreated individuals who are not indicated to have deceased by each census year.

Table I.16: Census Linkage Balance

	Older Allottees			Younger Allottees			All		
Variable	No Oil	Oil	p	No Oil	Oil	p	No Oil	Oil	p
Panel A: Dawes Rolls									
N	3,669	168		2,877	125		6,546	293	
Male	0.47 (0.50)	0.48 (0.50)	0.950	0.49 (0.50)	0.54 (0.50)	0.228	0.48 (0.50)	0.51 (0.50)	0.407
Age in 1910	36.86 (15.56)	35.97 (14.61)	0.445	10.70 (4.13)	10.99 (4.07)	0.440	25.37 (17.66)	25.31 (16.80)	0.953
Panel B: Matched to 1910 Census									
N	1,292	99		1,455	69		2,747	168	
Male	0.52 (0.50)	0.47 (0.50)	0.438	0.53 (0.50)	0.57 (0.50)	0.531	0.52 (0.50)	0.51 (0.50)	0.814
Age	36.65 (14.63)	34.36 (12.89)	0.096	10.08 (4.71)	11.26 (4.20)	0.026	22.57 (16.98)	24.88 (15.32)	0.062
Panel C: Matched to 1920 Census									
N	902	63		995	55		1,897	118	
Male	0.54 (0.50)	0.44 (0.50)	0.167	0.61 (0.49)	0.64 (0.49)	0.719	0.58 (0.49)	0.53 (0.50)	0.381
Age	44.77 (13.04)	42.78 (13.78)	0.268	18.94 (4.66)	20.38 (5.99)	0.084	31.22 (16.09)	32.34 (15.59)	0.453
Panel D: Matched to 1930 Census									
N	528	46		622	43		1,150	89	
Male	0.59 (0.49)	0.46 (0.50)	0.079	0.74 (0.44)	0.58 (0.50)	0.051	0.67 (0.47)	0.52 (0.50)	0.006
Age	51.86 (11.04)	52.11 (12.70)	0.897	29.40 (5.26)	30.28 (4.56)	0.233	39.71 (14.00)	41.56 (14.59)	0.250
Panel E: Matched to 1940 Census									
N	382	38		554	33		936	71	
Male	0.64 (0.48)	0.47 (0.51)	0.061	0.78 (0.41)	0.61 (0.50)	0.057	0.72 (0.45)	0.54 (0.50)	0.003
Age	59.91 (9.90)	59.16 (9.14)	0.632	39.40 (5.52)	40.70 (4.43)	0.116	47.77 (12.64)	50.58 (11.79)	0.058

Note: Balance test comparing pre-treatment characteristics of treated and untreated Creek Freedmen allottees by cohort (Older Allottees/Younger Allottees) and census year linkage status; p-values from two-sample t-tests shown.

Table I.17: Creek Allottee Summary Statistics

	Enrollment Category			
	Indian		Freedman	
	12,014	63.7%	6,836	36.3%
Age in 1910	26.8	(17.8)	25.3	(17.5)
Male	0.5	(0.5)	0.5	(0.5)

Note: Table provides counts (percents) and means (standard deviations) for variables from the Dawes Rolls (Campbell, 1915a,b).

Table I.18: Oil Discovery by Allottee Race

Group	Oil (1920)
Creek Indian	0.054 [N=12,014]
Freedman	0.052 [N=6,836]
p-value	0.454

Note: Mean (Standard deviation), p-value from a pooled t-test. Allottees in this table grouped by whether they were enrolled on the Dawes Roll as "Indian" or "Freedmen."

Table I.19: Balance on individual-level and household-level pre-treatment characteristics

Variable	No Oil	Oil	p-value
A. Location, 1910			
Urban	0.10 (0.30)	0.12 (0.32)	0.287
Living in Oklahoma	0.99 (0.09)	0.99 (0.08)	0.720
Farm	0.70 (0.46)	0.75 (0.43)	0.091
B. Family Status, 1910			
Household head	0.19 (0.40)	0.14 (0.34)	0.006
Never married/single (Under age 25)	0.91 (0.29)	0.98 (0.15)	0.000
Separated or divorced if ever married	0.08 (0.27)	0.03 (0.18)	0.020
Duration of current marriage, years	13.96 (11.25)	15.79 (10.97)	0.149
Children ever born	5.74 (3.81)	6.69 (3.48)	0.065
Children still living	3.84 (2.64)	5.19 (2.81)	0.001
C. Household Characteristics, 1910			
Household members	6.23 (2.66)	7.11 (2.43)	0.000
Household members under 18	3.53 (2.28)	4.33 (1.94)	0.000
Household members over 18	2.69 (1.24)	2.77 (1.28)	0.323
Child in school	0.85 (0.36)	0.89 (0.31)	0.094

Note: Mean (Standard deviation), p-value from a pooled t-test. Allottees in this table grouped by whether oil produced under own allotment by the end of 1918 or not.

Table I.20: Human Capital Outcomes for Young Allottees (Dry Well Comparison)

	Age 12 (1)	Age 14 (2)	Age 16 (3)	Age 18 (4)	Age 20 (5)	Age 22 (6)
Oil	-	-	0.089**	0.165	0.207	0.140
			(0.038)	(0.111)	(0.152)	(0.103)
Mean	1.00	0.98	0.92	0.62	0.25	0.11
Observations	195	168	122	88	72	83

Note: Sample restricted to Creek Freedmen allottees born after 1892 with either a producing or a dry well (comparison group) drilled on their own allotment by the end of 1918, matched to at least one census decade. Dependent variables are binary indicators for school enrollment at the indicated age, constructed via monotone inference from IPUMS enrollment records across the 1910–1930 censuses: an allottee observed in school at age X is coded as enrolled at all target ages $\leq X$; one observed not in school at age X is coded as not enrolled at all target ages $\geq X$. Completed years of education reported in the 1940 census extend the inference for individuals whose census observations fall outside their school-age years, assuming school entry at age 6. All specifications control for age in 1910 (linear and quadratic) and sex. Standard errors clustered by family in parentheses. Treatment is defined as the presence of a producing oil well on own allotment by the end of 1918. For more information on data and variable definitions, see Data Appendix H. Dashes mark cells where the outcome does not vary among treated allottees at the indicated age; the linear probability model is uninformative at that boundary, so the coefficient and standard error are suppressed while the mean and sample size are retained. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$

Table I.21: The Effect of Oil Money on Labor Force Entry and Household Formation for Young Allottees (Dry Well Comparison)

Outcome	Age 16 (1)	Age 18 (2)	Age 20 (3)	Age 22 (4)	Age 24 (5)	Age 26 (6)	Age 28 (7)	Age 30 (8)
Panel A: Men								
In school	-	-0.044	0.098	0.140	-	-	-	-
		(0.163)	(0.195)	(0.131)				
Mean	0.91	0.65	0.26	0.13	0.03	0.02	0.01	0.01
N	69	51	42	54	68	89	103	108
In workforce	-0.022	-0.093	-0.137	-0.108	-0.074	-0.047	-0.036	-
	(0.134)	(0.128)	(0.116)	(0.107)	(0.076)	(0.055)	(0.036)	
Mean	0.29	0.58	0.77	0.86	0.93	0.96	0.99	1.00
N	62	64	66	73	82	95	102	105
Ever married	-	-	-0.045	-0.101	0.158	0.106	0.109	0.023
			(0.103)	(0.149)	(0.142)	(0.104)	(0.068)	(0.054)
Mean	0.02	0.03	0.16	0.32	0.55	0.72	0.85	0.92
N	95	65	50	41	49	65	73	75
Head/spouse	-	-	-0.032	-0.128	0.049	-0.006	0.098	0.040
			(0.061)	(0.079)	(0.117)	(0.134)	(0.118)	(0.116)
Mean	0.02	0.02	0.08	0.16	0.30	0.45	0.61	0.69
N	111	86	72	63	69	71	74	72
Panel B: Women								
In school	0.021	0.372**	0.368	0.146	-	-	-	-
	(0.070)	(0.138)	(0.233)	(0.165)				
Mean	0.92	0.59	0.23	0.07	0.03	0.02	0.02	0.02
N	53	37	30	29	39	55	63	65
In workforce	-0.001	-0.069	-0.080	-0.052	0.051	-0.039	-0.048	-0.045
	(0.074)	(0.094)	(0.126)	(0.170)	(0.191)	(0.200)	(0.176)	(0.113)
Mean	0.10	0.22	0.29	0.35	0.41	0.59	0.76	0.93
N	83	72	55	48	44	39	34	29
Ever married	-	-0.047	-0.147	-0.048	0.021	0.064	-	-
		(0.098)	(0.230)	(0.300)	(0.218)	(0.108)		
Mean	0.02	0.12	0.37	0.56	0.72	0.86	0.96	0.96
N	61	41	27	25	32	43	45	48
Head/spouse	-	-	-0.068	0.011	-0.062	-0.173	0.016	-0.016
			(0.105)	(0.147)	(0.162)	(0.167)	(0.167)	(0.156)
Mean	0.01	0.03	0.12	0.21	0.38	0.53	0.72	0.76
N	77	59	41	34	37	40	39	41

Note: Sample restricted to Creek Freedmen allottees born after 1892 with either a producing or a dry well (comparison group) drilled on their own allotment by the end of 1918, matched to at least one census decade. Dependent variables are binary indicators for the outcome listed, measured at the indicated age via monotone inference across the 1910–1940 censuses. Specifications are estimated separately by gender and control for age in 1910 (linear and quadratic). Standard errors clustered by family in parentheses. Treatment is defined as the presence of a producing oil well on own allotment by the end of 1918. For more information on data and variable definitions, see Data Appendix H. Dashes mark cells where the outcome does not vary among treated allottees at the indicated age; the linear probability model is uninformative at that boundary, so the coefficient and standard error are suppressed while the mean and sample size are retained. ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$

Table I.22: Fertility (Dry Well Comparison)

	Women (Oil) (1)	Men (Oil) (2)	Pooled (Oil) (3)
Panel A: Offspring count			
By age 40	0.560	-0.387	0.107
	(0.668)	(0.423)	(0.383)
Mean	3.652	2.321	2.919
N	89	109	198
By age 50	1.009	-0.465	0.237
	(1.012)	(0.673)	(0.581)
Mean	4.354	2.959	3.512
N	48	73	121
Panel B: School attendance, ages 6–17			
	-0.003	0.011	0.057*
	(0.036)	(0.061)	(0.030)
Mean	0.905	0.803	0.844
N	327	264	752

Note: Each coefficient in the fertility panels is from an OLS regression of the outcome on whether the allottee was treated as defined in the Data Appendix, controlling for age in 1910, age in 1910 squared, and sex (pooled column only). The sample is restricted to Creek Freedmen allottees with either a producing oil well (treated) or a dry well (comparison group) drilled on their own allotment by the end of 1918. Number of offspring by age x counts children observed in census records linked to the allottee. For the school attendance panel, the Mothers (Fathers) column reports the effect of the mother's (father's) treatment on the child's probability of school attendance; the Pooled column uses an indicator for treatment of either parent. School attendance controls include child age, child age squared, child sex, and census year fixed effects; the sample is children of Creek allottees ages 6–17 observed in 1920, 1930, or 1940. Standard errors clustered by Family ID. *** $p < .01$, ** $p < .05$, * $p < .1$

Table I.23: The Effect of Oil Money on Migration (Dry Well Comparison)

	(1)	(2)
Panel A: Left County		
Coefficient	-0.054	-0.065
Std. error	(0.049)	(0.044)
Mean	0.62	0.62
Observations	873	873
Panel B: Left Creek Nation		
Coefficient	-0.025	-0.030
Std. error	(0.036)	(0.036)
Mean	0.21	0.21
Observations	873	873
Panel C: Left Oklahoma		
Coefficient	-0.002	-0.005
Std. error	(0.033)	(0.034)
Mean	0.15	0.15
Observations	873	873
County FE?	No	Yes

Note: Sample restricted to Creek Freedmen allottees with either a producing oil well (treated) or a dry well (comparison group) drilled on their own allotment by the end of 1918. This table presents regression estimates of the difference in likelihood of being observed out of Oklahoma, out of the Muscogee (Creek) Nation, or out of allotment county, respectively, between treated and comparison allottees in 1920, 1930, and 1940. I define allotment county as the Oklahoma county in which the allottee's Dawes Enrollment was located. Regressions control for age, age squared, and gender and include fixed effects for census decade. Treatment is defined as the presence of a producing oil well on own allotment by the end of 1918. For more information on data and variable definitions, see Data Appendix H. Standard errors are clustered at the Family ID level and are shown in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$

Table I.24: The Effect of Oil Money on Urbanization (Dry Well Comparison)

	Older Allottees				Younger Allottees				Pooled			
	Urban	Stayer	Urban(S)	Urban(M)	Urban	Stayer	Urban(S)	Urban(M)	Urban	Stayer	Urban(S)	Urban(M)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Oil	0.071	-0.151***	-0.001	0.119	0.133	-0.091	0.070	0.139	0.101*	-0.123***	0.030	0.131
	(0.068)	(0.047)	(0.057)	(0.107)	(0.089)	(0.056)	(0.126)	(0.118)	(0.058)	(0.037)	(0.067)	(0.083)
Mean(Y)	0.211	0.656	0.077	0.330	0.342	0.582	0.235	0.420	0.276	0.624	0.151	0.377
N	194	633	91	103	193	478	81	112	387	1,111	172	215

Note: Sample restricted to Creek Freedmen allottees with either a producing oil well (treated) or a dry well (comparison group) drilled on their own allotment by the end of 1918. This table presents regression estimates of the effect of oil wealth on urbanization and related outcomes. Outcomes include: living in an urban area in 1920, remaining in one's 1910 census township through 1920 (Stayer), 1920 urban residence conditional on remaining (Urban(S)), and 1920 urban residence conditional on moving (Urban(M)). The sample comprises allottees matched in both the 1910 and 1920 censuses. Specifications are estimated separately for Older Allottees and Younger Allottees, and pooled. Treatment is defined as the presence of a producing oil well on own allotment by the end of 1918. Standard errors clustered at the Family ID level are shown in parentheses. ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$

Table I.25: The Effect of Oil Money on Homeownership, Older Allottees (Dry Well Comparison)

	Panel A: 1920		Panel B: 1930		Panel C: 1940		Panel D: 1920–1940 Pooled	
	Homeowner	Free and Clear	Homeowner	House Value	Homeowner	House Value	Homeowner	House Value
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Oil	-0.091	-0.036	-0.012	666.740	0.160	827.438	-0.005	836.824
	(0.075)	(0.099)	(0.093)	(1917.626)	(0.105)	(825.461)	(0.063)	(1035.848)
Mean	0.60	0.63	0.52	1969.55	0.47	930.59	0.54	1266.72
N	194	120	131	22	103	46	428	68

Note: Sample restricted to Creek Freedmen allottees with either a producing oil well (treated) or a dry well (comparison group) drilled on their own allotment by the end of 1918, born in 1892 or earlier (Older Allottees). This table presents regression estimates of the differences in homeownership between treated and comparison allottees in 1920, 1930, and 1940, and pooled across 1920–1940. All regressions control for age, age squared, and gender, and the pooled regressions control for year. Treatment is defined as the presence of a producing oil well on own allotment by the end of 1918. For more information on data and variable definitions, see Data Appendix H. Standard errors are clustered by family groups and are shown in parentheses. ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$

Table I.26: The Effect of Oil Money on Homeownership, Younger Allottees (Dry Well Comparison)

	Panel A: 1930		Panel B: 1940		Panel C: 1920–1940 Pooled	
	Homeowner (1)	House Value (2)	Homeowner (3)	House Value (4)	Homeowner (5)	House Value (6)
Oil	0.067 (0.081)	4586.651 (4104.666)	0.044 (0.089)	-445.491 (2413.369)	0.037 (0.054)	1142.361 (2555.034)
Mean	0.26	5020.31	0.23	1966.67	0.23	3102.91
N	134	16	118	27	378	43

Note: Sample restricted to Creek Freedmen allottees with either a producing oil well (treated) or a dry well (comparison group) drilled on their own allotment by the end of 1918, and born after 1892. This table presents regression estimates of the differences in homeownership between treated and comparison allottees in 1930 and 1940, and pooled across 1920–1940. All regressions control for age, age squared, and gender, and the pooled regressions control for year. Treatment is defined as the presence of a producing oil well on own allotment by the end of 1918. For more information on data and variable definitions, see Data Appendix H. Standard errors are clustered by family groups and are shown in parentheses. ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$

Table I.27: Neighborhood Quality, 1940 (Dry Well Comparison)

Outcome	Estimate	Mean	N
Panel A: Education			
Share graduated college	0.072*	0.14	221
	(0.042)		
Share graduated high school	0.071*	0.26	221
	(0.038)		
School attendance	0.018*	0.83	221
	(0.009)		
Teacher Salary	89.219*	933.96	204
	(52.975)		
Panel B: Labor Markets			
Employment to population ratio	0.014	0.80	221
	(0.018)		
Unemployment rate	-0.014	0.14	221
	(0.016)		
Racial earnings gap	32.325	445.36	183
	(50.567)		
Panel C: Families			
Share of families single-parent	0.008	0.18	221
	(0.015)		
Poverty rate	-0.017	0.72	221
	(0.027)		
Panel D: Segregation			
Proportion of neighbors white	-0.034	0.54	221
	(0.052)		
Panel E: Other			
Urban	-0.007	0.43	221
	(0.071)		

Note: Sample restricted to Creek Freedmen allottees with either a producing oil well (treated) or a dry well (comparison group) drilled on own allotment by the end of 1918. This table presents regression estimates of selected neighborhood characteristics of the places where treated and comparison allottees lived in 1940. All regressions control for age, age squared, and gender. Treatment is defined as the presence of a producing oil well by the end of 1918 on allotment. For more information on data and variable definitions, see Data Appendix H. Standard errors are clustered by Family ID and are shown in parentheses. ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$

Table I.28: The Effect of Oil Money on Occupational Status: Multinomial Logit Results (No Handlinks)

	Older Allottees		Younger Allottees		Pooled	
	White Collar	Farming	White Collar	Farming	White Collar	Farming
	(1)	(2)	(3)	(4)	(5)	(6)
Oil	0.962	-0.362	1.224**	-0.423	1.047**	-0.384
	(0.673)	(0.432)	(0.478)	(0.585)	(0.422)	(0.353)
N	1391	1391	1255	1255	2646	2646

Note: The sample excludes all individuals linked to any census year by hand. Each column presents a multinomial logit equation for the named occupational outcome relative to blue-collar employment, which serves as the base category. Panel A: Reported coefficients are log-odds ratios. Panel B: average marginal effects. Treatment is defined as the presence of a producing oil well on the allottee's own allotment by the end of 1918. All specifications include year fixed effects and control for age in 1910, age in 1910 squared, and an indicator for male. Standard errors clustered at the Family ID level are reported in parentheses. *$p < 0.10$, **$p < 0.05$, ***$p < 0.01$.

Table I.29: The Effect of Oil Money on Labor Force Entry and Household Formation for Younger Allottees (No Handlinks)

Outcome	Age 16 (1)	Age 18 (2)	Age 20 (3)	Age 22 (4)	Age 24 (5)	Age 26 (6)	Age 28 (7)	Age 30 (8)
Panel A: Men								
In school	-	0.034	0.150	0.211	-	-	-	-
		(0.214)	(0.214)	(0.213)				
Mean	0.90	0.59	0.19	0.10	0.04	0.03	0.01	0.01
N	330	240	179	206	267	393	477	498
In workforce	-0.031	-0.093	-0.077	-0.111	-0.090	-0.034	-	-
	(0.147)	(0.147)	(0.129)	(0.134)	(0.107)	(0.072)		
Mean	0.36	0.68	0.85	0.90	0.94	0.96	0.99	1.00
N	313	323	309	321	365	456	500	516
Ever married	-	-	-	-	0.292	0.133	0.103	0.068
					(0.180)	(0.131)	(0.084)	(0.072)
Mean	0.00	0.02	0.06	0.14	0.31	0.60	0.80	0.86
N	469	330	226	207	225	272	316	333
Head/spouse	-	-	-	-	0.176	0.086	0.138	0.086
					(0.126)	(0.133)	(0.132)	(0.128)
Mean	0.00	0.01	0.03	0.05	0.14	0.30	0.52	0.60
N	546	437	353	329	336	323	325	316
Panel B: Women								
In school	-	-	-	-	-	-	-	-
Mean	0.86	0.54	-	-	-	-	0.01	0.01
N	229	145	-	-	-	-	228	231
Ever married	-	-	-	-	-	-	-	-
Mean	-	-	-	-	-	-	-	-
N	-	-	-	-	-	-	-	-
Head/spouse	-	-	-	-	-	-	-	-
Mean	-	-	-	-	-	-	-	-
N	-	-	-	-	-	-	-	-

Note: The sample excludes all individuals linked to any census year by hand. Sample is Creek Freedmen allottees born after 1892, matched to at least one census decade. Dependent variables are binary indicators for the outcome listed, measured at the indicated age via monotone inference across the 1910–1940 censuses. Specifications are estimated separately by gender and control for age in 1910 (linear and quadratic). Standard errors clustered by family in parentheses. Treatment is defined as the presence of a producing oil well on own allotment by the end of 1918. For more information on data and variable definitions, see Data Appendix H. Dashes mark cells where the outcome does not vary among treated allottees at the indicated age; the linear probability model is uninformative at that boundary, so the coefficient and standard error are suppressed while the mean and sample size are retained. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$

Table I.30: Fertility (No Handlinks)

	Women (Oil) (1)	Men (Oil) (2)	Pooled (Oil) (3)
Panel A: Offspring count			
By age 40	0.414	-1.014***	-0.563*
	(0.599)	(0.320)	(0.329)
Mean	2.968	2.559	2.728
N	317	449	766
By age 50	0.471	-0.910	-0.363
	(0.590)	(0.711)	(0.535)
Mean	3.330	3.355	3.345
N	209	290	499
Panel B: School attendance, ages 6–17			
	-0.151	0.037	0.047
	(0.100)	(0.056)	(0.043)
Mean	0.832	0.825	0.825
N	819	1,123	1,853

Note: The sample excludes all individuals linked to any census year by hand. Each coefficient in the fertility panels is from an OLS regression of the outcome on an indicator for whether the allottee was treated as defined in the Data Appendix, controlling for age in 1910, age in 1910 squared, and sex (pooled column only). The sample is Creek Freedmen allottees. Number of offspring by age x counts children observed in census records linked to the allottee. The numbers reflect a substantial effort at deduplication, but instances of individuals observed in different decades but counted as multiple offspring likely remain. For the school attendance panel, the Mothers (Fathers) column reports the effect of the mother's (father's) treatment on the child's probability of school attendance; the Pooled column uses an indicator for treatment of either parent. School attendance controls include child age, child age squared, child sex, and census year fixed effects; the sample is children of Creek allottees ages 6–17 observed in 1920, 1930, or 1940. Standard errors clustered by Family ID. *** $p < .01$, ** $p < .05$, * $p < .1$

Table I.31: The Effect of Oil Money on Migration (No Handlinks)

	(1)	(2)
Panel A: Left County		
Coefficient	-0.098	-0.092
Std. error	(0.061)	(0.057)
Mean	0.66	0.66
Observations	3,574	3,574
Panel B: Left Creek Nation		
Coefficient	-0.051	-0.023
Std. error	(0.051)	(0.052)
Mean	0.25	0.25
Observations	3,574	3,574
Panel C: Left Oklahoma		
Coefficient	0.045	0.044
Std. error	(0.050)	(0.052)
Mean	0.14	0.14
Observations	3,574	3,574
County FE?	No	Yes

Note: The sample excludes all individuals linked to any census year by hand. This table presents regression estimates of the difference in likelihood of leaving the allotment county, leaving the Creek Nation, and leaving Oklahoma between treated and untreated Creek Freedmen allottees, pooling the 1920, 1930, and 1940 censuses with year fixed effects. All regressions control for age, age squared, and gender. Treatment is defined as the presence of a producing oil well on own allotment by the end of 1918. For more information on data and variable definitions, see Data Appendix H. Standard errors are clustered at the Family ID level and are shown in parentheses. $***p < 0.01$, $**p < 0.05$, $*p < 0.1$

Table I.32: The Effect of Oil Money on Urbanization (No Handlinks)

	Older Allottees				Younger Allottees				Pooled			
	Urban	Stayer	Urban(S)	Urban(M)	Urban	Stayer	Urban(S)	Urban(M)	Urban	Stayer	Urban(S)	Urban(M)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Oil	0.076	-0.008	-	0.268*	0.256**	0.051	0.103	0.264*	0.152*	0.016	-0.048	0.265**
	(0.086)	(0.046)		(0.145)	(0.119)	(0.054)	(0.179)	(0.136)	(0.080)	(0.038)	(0.056)	(0.110)
Mean(Y)	0.190	0.742	0.113	0.264	0.234	0.645	0.139	0.323	0.212	0.700	0.126	0.294
N	863	3,663	424	439	875	2,741	423	452	1,738	6,404	847	891

Note: The sample excludes all individuals linked to any census year by hand. This table presents regression estimates of the effect of oil wealth on urbanization and related outcomes. Outcomes include: living in an urban area in 1920, remaining in one's 1910 census township through 1920 (Stayer), 1920 urban residence conditional on remaining (Urban(S)), and 1920 urban residence conditional on moving (Urban(M)). The sample comprises allottees matched in both the 1910 and 1920 censuses. Specifications are estimated separately for Older Allottees and Younger Allottees, and pooled. Treatment is defined as the presence of a producing oil well on own allotment by the end of 1918. Standard errors clustered at the Family ID level are shown in parentheses. Dashes mark cells where the outcome does not vary among treated allottees; the linear probability model is uninformative at that boundary, so the coefficient and standard error are suppressed while the mean and sample size are retained. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$

Table I.33: The Effect of Oil Money on Homeownership, Older Allottees (No Handlinks)

	Panel A: 1920		Panel B: 1930		Panel C: 1940		Panel D: 1920–1940 Pooled	
	Homeowner	Free and Clear	Homeowner	House Value	Homeowner	House Value	Homeowner	House Value
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Oil	-0.001	-0.089	0.103	2066.645	0.210	1774.303	0.073	1843.895
	(0.104)	(0.128)	(0.110)	(3203.414)	(0.139)	(2042.443)	(0.091)	(2295.047)
Mean	0.56	0.66	0.48	1663.91	0.44	925.07	0.51	1157.67
N	862	542	499	68	358	148	1719	216

Note: The sample excludes all individuals linked to any census year by hand. This table presents regression estimates of the differences in homeownership between treated and untreated Creek Freedmen allottees born in 1892 or earlier (Older Allottees) in 1920, 1930, and 1940, and pooled across 1920–1940. All regressions control for age, age squared, and gender, and the pooled regressions control for year. Treatment is defined as the presence of a producing oil well on own allotment by the end of 1918. For more information on data and variable definitions, see Data Appendix H. Standard errors are clustered by family groups and are shown in parentheses. ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$

Table I.34: The Effect of Oil Money on Homeownership, Younger Allottees (No Handlinks)

	Panel A: 1930		Panel B: 1940		Panel C: 1920–1940 Pooled	
	Homeowner (1)	House Value (2)	Homeowner (3)	House Value (4)	Homeowner (5)	House Value (6)
Oil	0.002 (0.096)	2772.967 (1869.398)	0.063 (0.121)	2473.477 (1989.541)	0.059 (0.073)	2302.557* (1372.455)
Mean	0.19	2526.30	0.22	1363.32	0.18	1761.60
N	516	50	460	96	1422	146

Note: The sample excludes all individuals linked to any census year by hand. This table presents regression estimates of the differences in homeownership between treated and untreated Creek Freedmen allottees born after 1892 (Younger Allottees) in 1930 and 1940, and pooled across 1920–1940. All regressions control for age, age squared, and gender, and the pooled regressions control for year. Treatment is defined as the presence of a producing oil well on own allotment by the end of 1918. For more information on data and variable definitions, see Data Appendix H. Standard errors are clustered by family groups and are shown in parentheses. ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$

Table I.35: Neighborhood Quality, 1940 (No Handlinks)

Outcome	Estimate	Mean	N
Panel A: Education			
Share graduated college	0.008	0.14	817
	(0.059)		
Share graduated high school	0.002	0.28	817
	(0.052)		
School attendance	0.024**	0.83	817
	(0.011)		
Teacher Salary	54.537	942.70	759
	(71.586)		
Panel B: Labor Markets			
Employment to population ratio	-0.023	0.81	817
	(0.028)		
Unemployment rate	0.011	0.13	816
	(0.019)		
Racial earnings gap	-31.517	487.41	671
	(59.022)		
Panel C: Families			
Share of families single-parent	0.011	0.17	815
	(0.020)		
Poverty rate	0.002	0.70	817
	(0.035)		
Panel D: Segregation			
Proportion of neighbors white	-0.022	0.59	817
	(0.066)		
Panel E: Other			
Urban	0.036	0.44	820
	(0.094)		

Note: The sample excludes all individuals linked to any census year by hand. This table presents regression estimates of selected neighborhood characteristics of the places where treated and untreated Creek Freedmen allottees lived in 1940. All regressions control for age, age squared, and gender. Treatment is defined as the presence of a producing oil well by the end of 1918 on own allotment. For more information on data and variable definitions, see Data Appendix H. Standard errors are clustered at the Family ID level and are shown in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$